

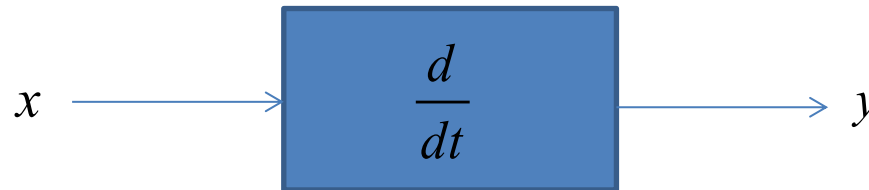
BDA30703

Block Diagram Representation of Control Systems

Hanani Abd Wahab

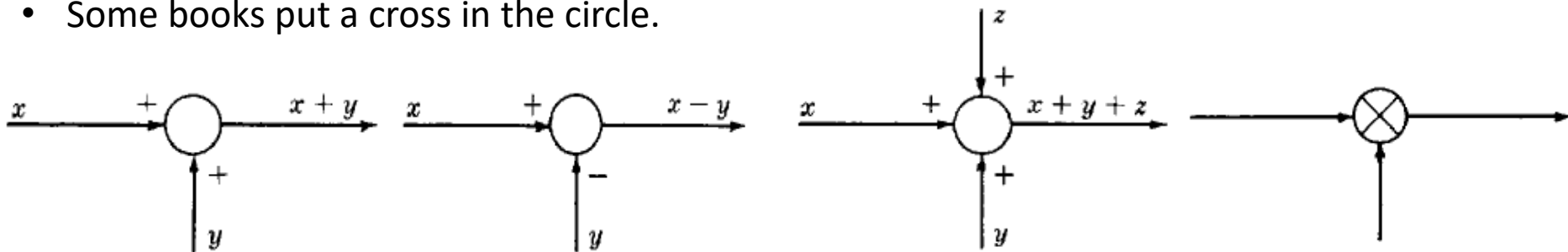
Introduction

- A Block Diagram is a shorthand pictorial representation of the cause-and-effect relationship of a system.
- The interior of the rectangle representing the block usually contains a description of or the name of the element, gain, or the symbol for the mathematical operation to be performed on the input to yield the output.
- The arrows represent the direction of information or signal flow.



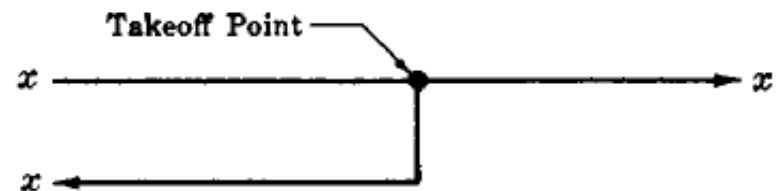
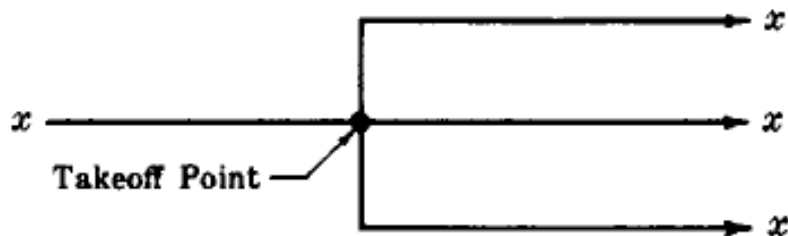
Introduction

- The operations of addition and subtraction have a special representation.
- The block becomes a small circle, called a summing point, with the appropriate plus or minus sign associated with the arrows entering the circle.
- The output is the algebraic sum of the inputs.
- Any number of inputs may enter a summing point.
- Some books put a cross in the circle.



Introduction

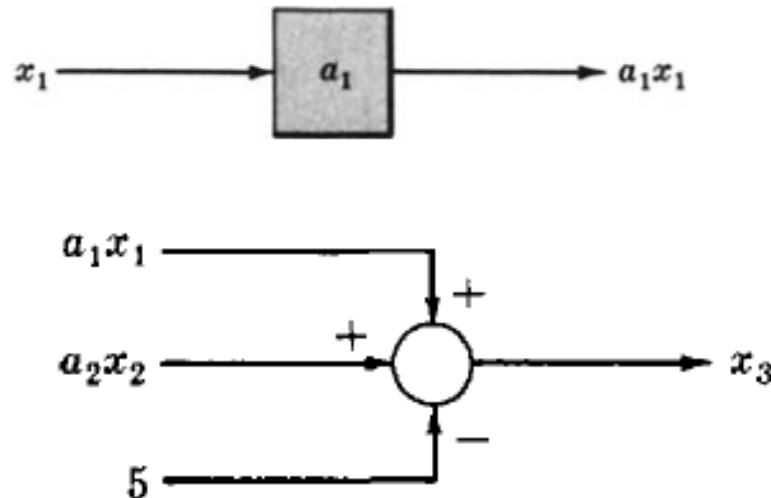
- In order to have the same signal or variable be an input to more than one block or summing point, a takeoff (or pickoff) point is used.
- This permits the signal to proceed unaltered along several different paths to several destinations.



Example-1

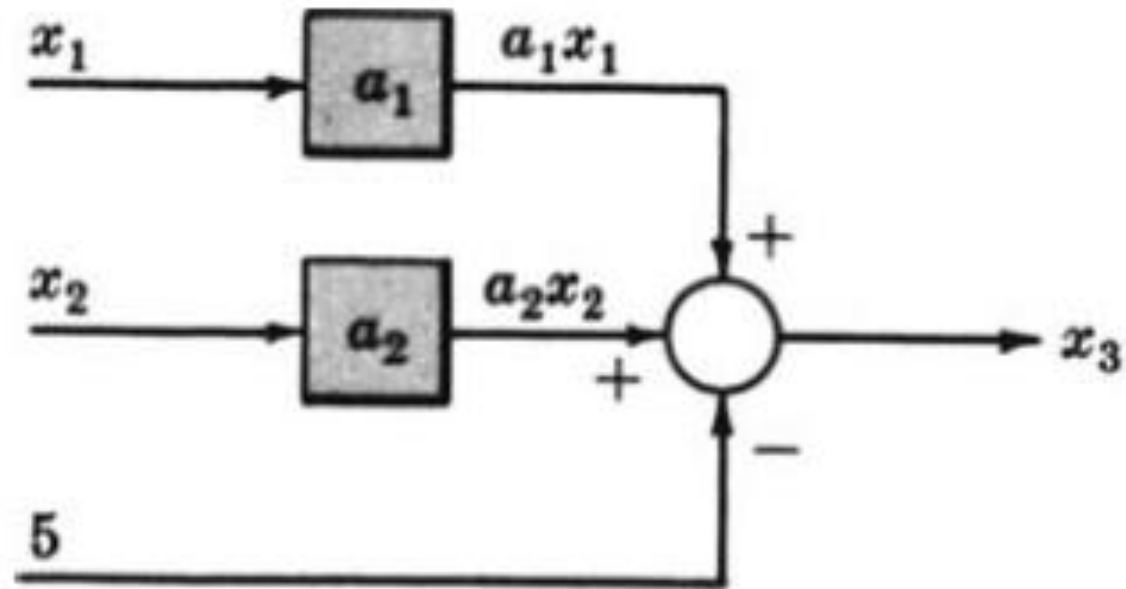
- Consider the following equations in which x_1 , x_2 , x_3 , are variables, and a_1 , a_2 are general coefficients or mathematical operators.

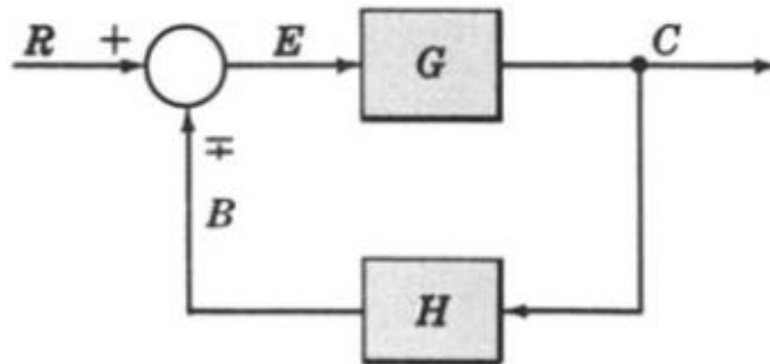
$$x_3 = a_1x_1 + a_2x_2 - 5$$



Example-1

$$x_3 = a_1x_1 + a_2x_2 - 5$$





Canonical Form of A Feedback Control System

$G \equiv$ direct transfer function $\equiv$ forward transfer function

$H \equiv$ feedback transfer function

$GH \equiv$ loop transfer function $\equiv$ open-loop transfer function

$C/R \equiv$ closed-loop transfer function $\equiv$ control ratio $\frac{C}{R} = \frac{G}{1 \pm GH}$

$E/R \equiv$ actuating signal ratio $\equiv$ error ratio $\frac{E}{R} = \frac{1}{1 \pm GH}$

$B/R \equiv$ primary feedback ratio $\frac{B}{R} = \frac{GH}{1 \pm GH}$

Characteristic Equation

- The control ratio is the closed loop transfer function of the system.

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 \pm G(s)H(s)}$$

- The denominator of closed loop transfer function determines the characteristic equation of the system.
- Which is usually determined as:

$$1 \pm G(s)H(s) = 0$$

Example-3

1. Open loop transfer function $\frac{B(s)}{E(s)} = G(s)H(s)$

2. Feed Forward Transfer function $\frac{C(s)}{E(s)} = G(s)$

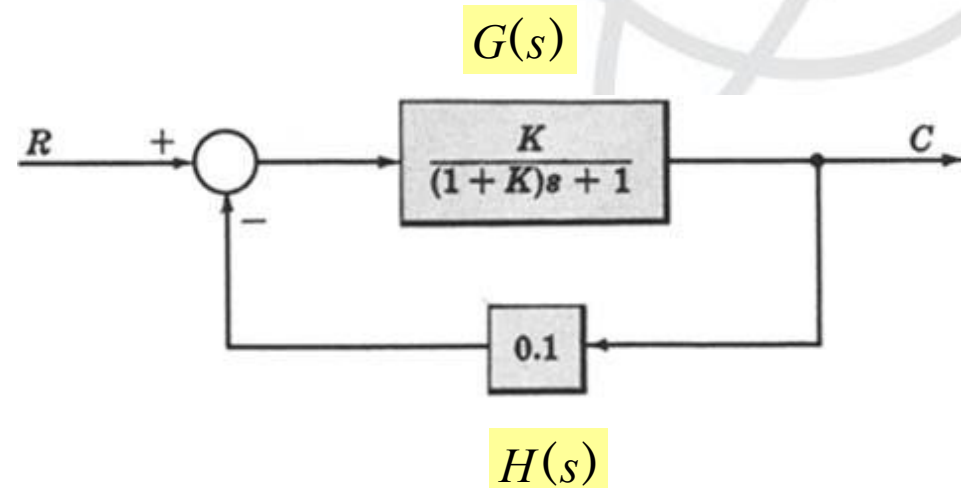
3. control ratio $\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$

4. feedback ratio $\frac{B(s)}{R(s)} = \frac{G(s)H(s)}{1 + G(s)H(s)}$

5. error ratio $\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)}$

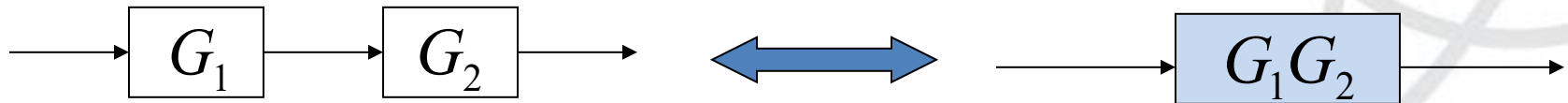
6. closed loop transfer function $\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$

7. characteristic equation $1 + G(s)H(s) = 0$

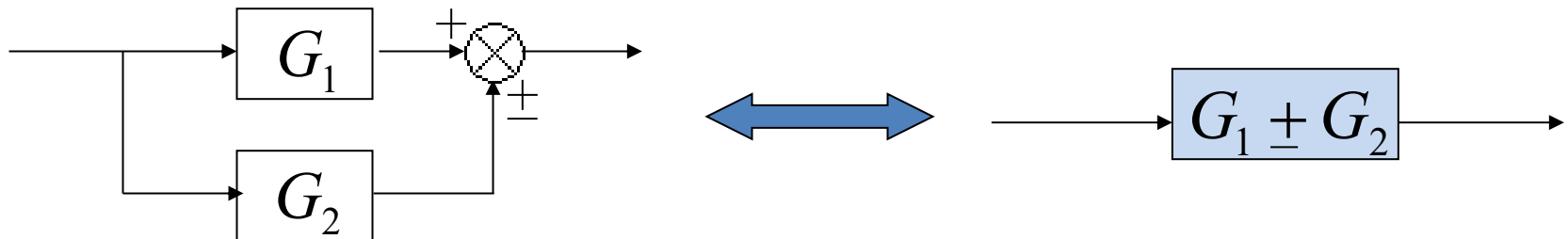


Reduction techniques

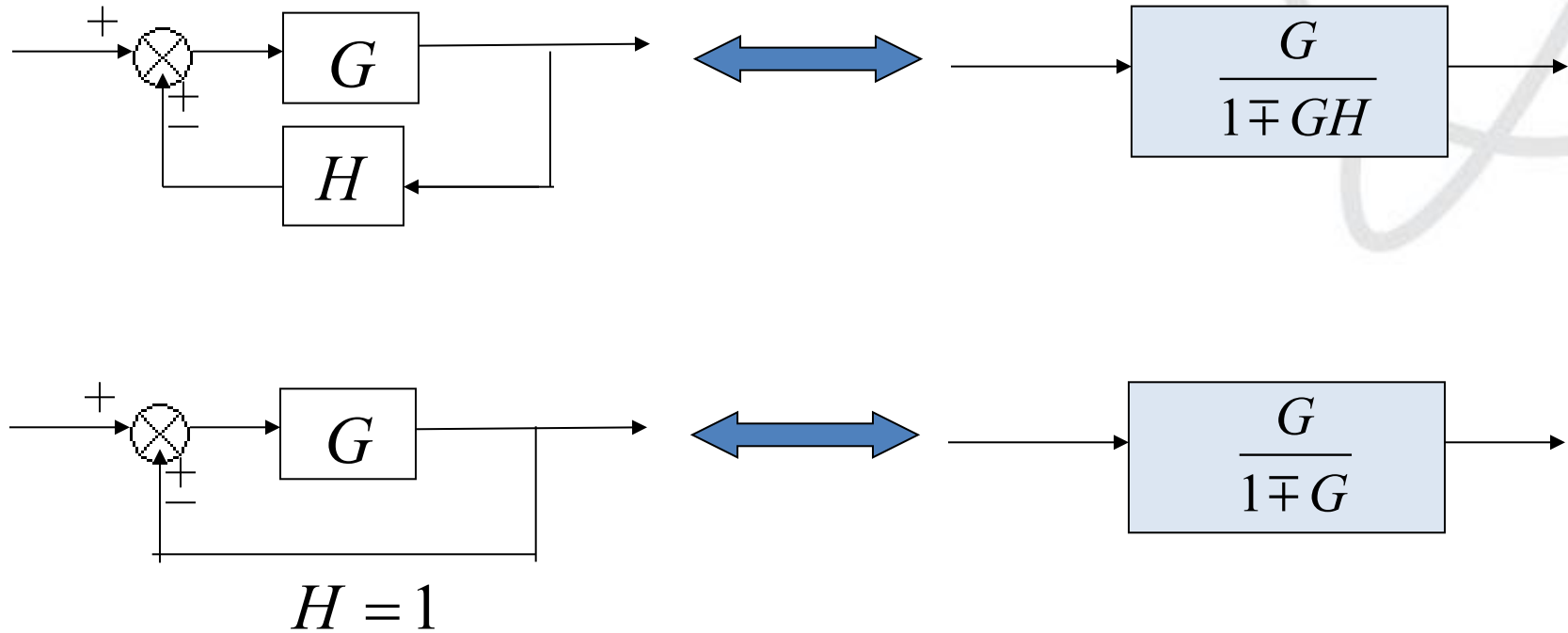
1. Combining blocks in cascade



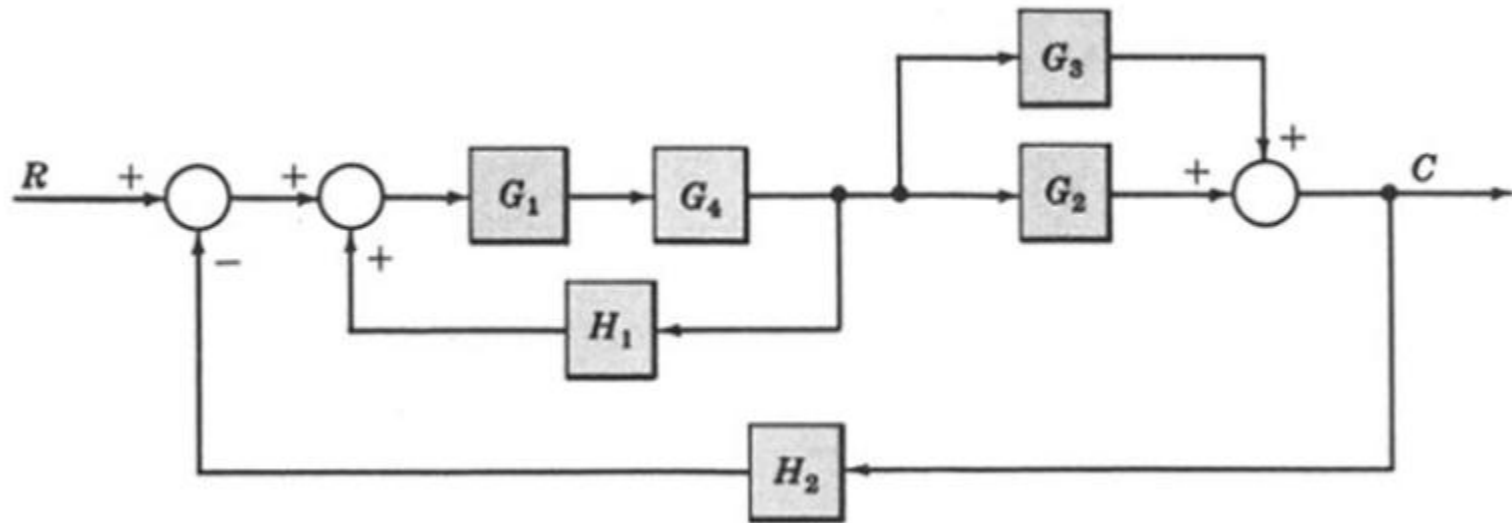
2. Combining blocks in parallel



3. Eliminating a feedback loop



Example-4: Reduce the Block Diagram to Canonical Form.



Step 1: Combine all cascade blocks

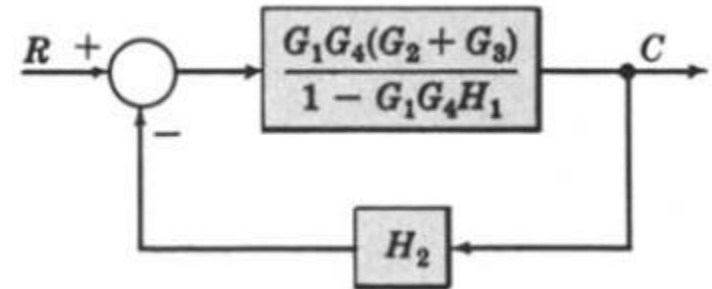
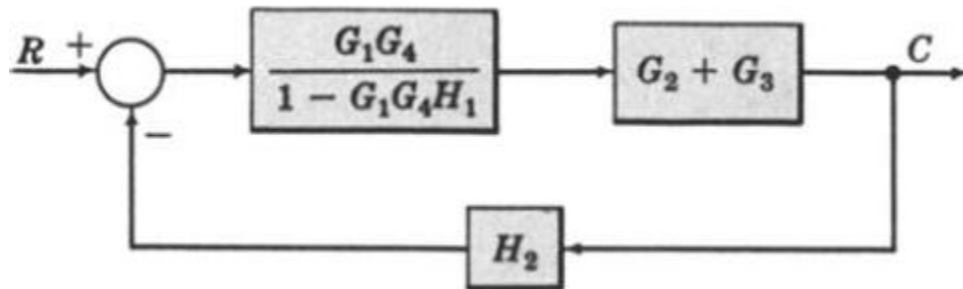


Step 2: Combine all parallel blocks



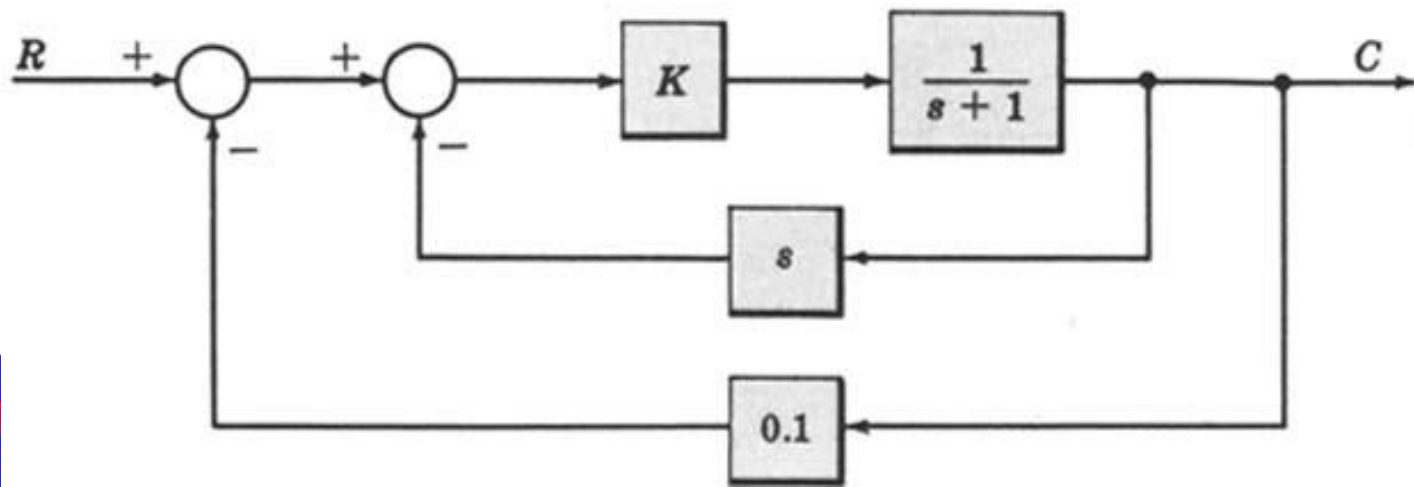
Example-4: Continue.

Step 3: Eliminate all minor feedback loops



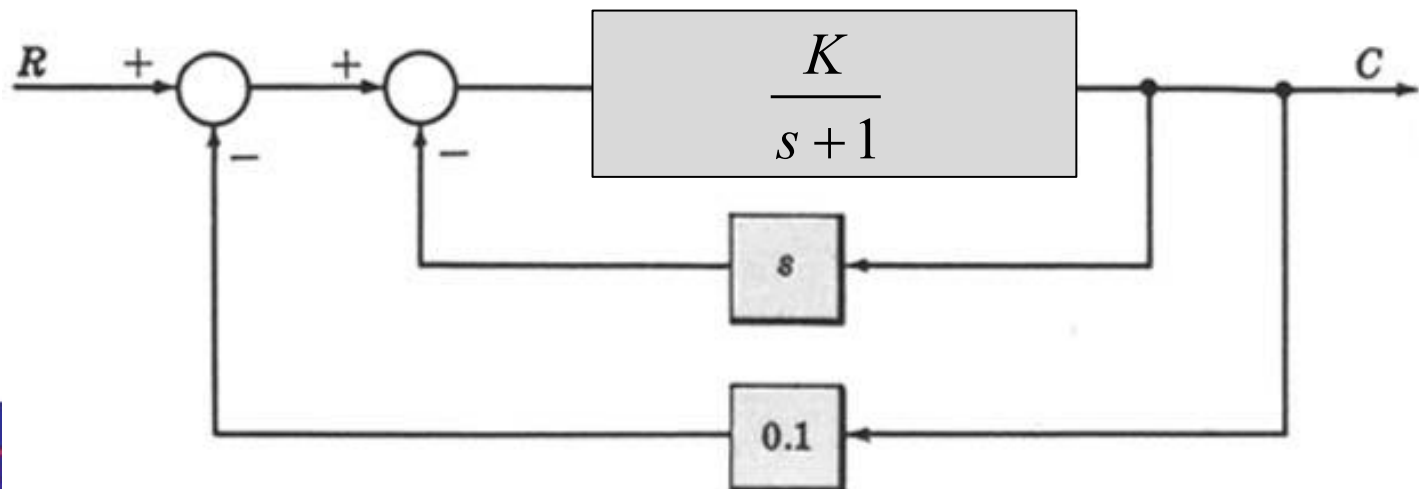
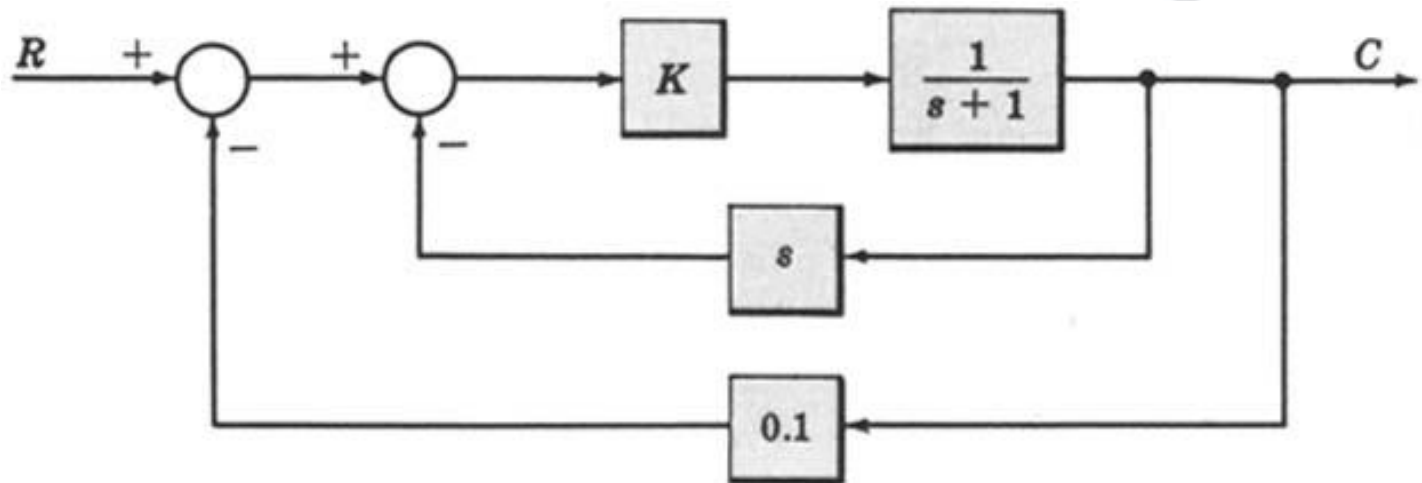
Example-5

- For the system represented by the following block diagram determine:
 - Open loop transfer function
 - Feed Forward Transfer function
 - control ratio
 - feedback ratio
 - error ratio
 - closed loop transfer function
 - characteristic equation
 - closed loop poles and zeros if $K=10$.

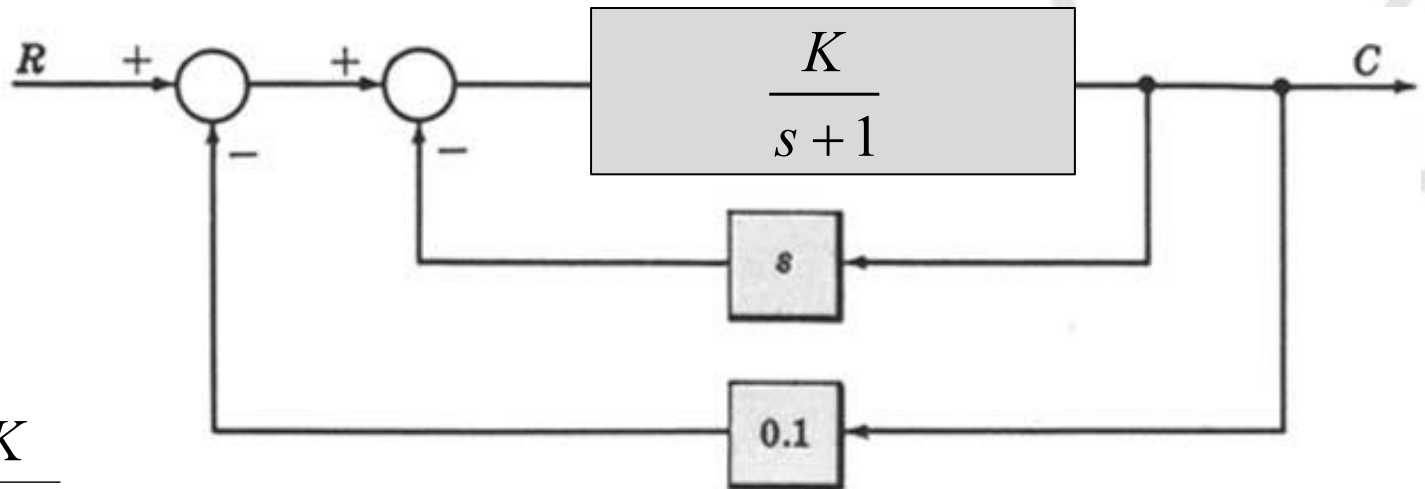


Example-5

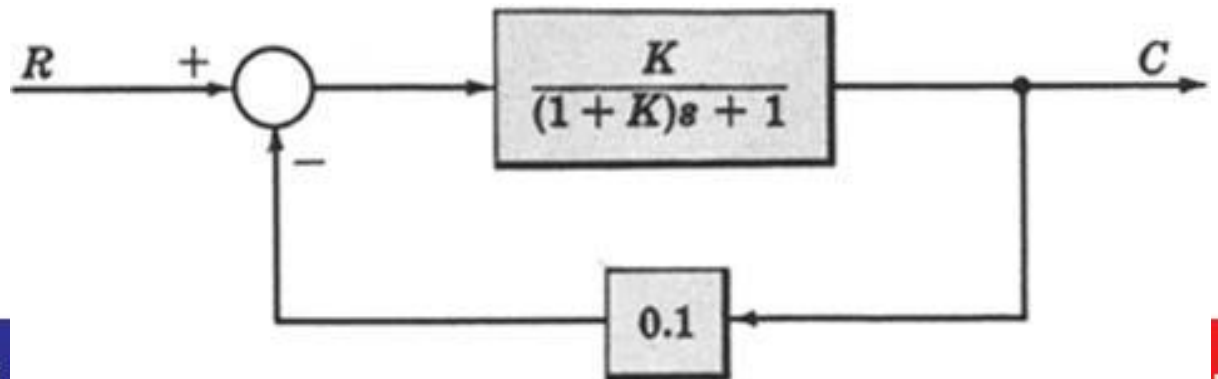
- First we will reduce the given block diagram to canonical form



Example-5



$$\frac{G}{1+GH} = \frac{\frac{K}{s+1}}{1 + \frac{K}{s+1}s}$$



Example-5 (see example-3)

1. Open loop transfer function $\frac{B(s)}{E(s)} = G(s)H(s)$

2. Feed Forward Transfer function $\frac{C(s)}{E(s)} = G(s)$

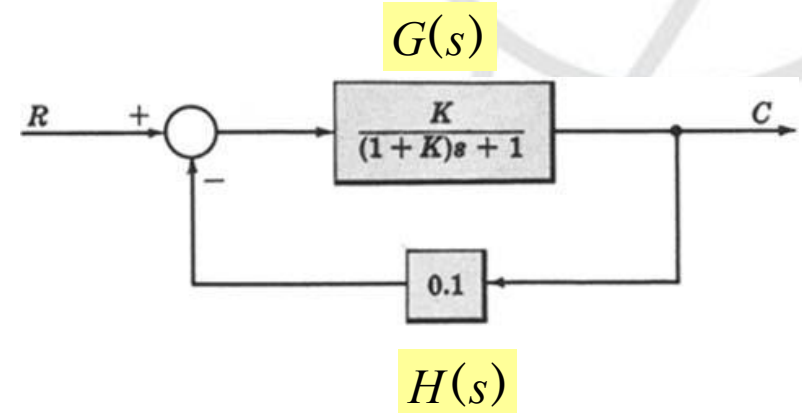
3. control ratio $\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$

4. feedback ratio $\frac{B(s)}{R(s)} = \frac{G(s)H(s)}{1 + G(s)H(s)}$

5. error ratio $\frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)}$

6. closed loop transfer function $\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$

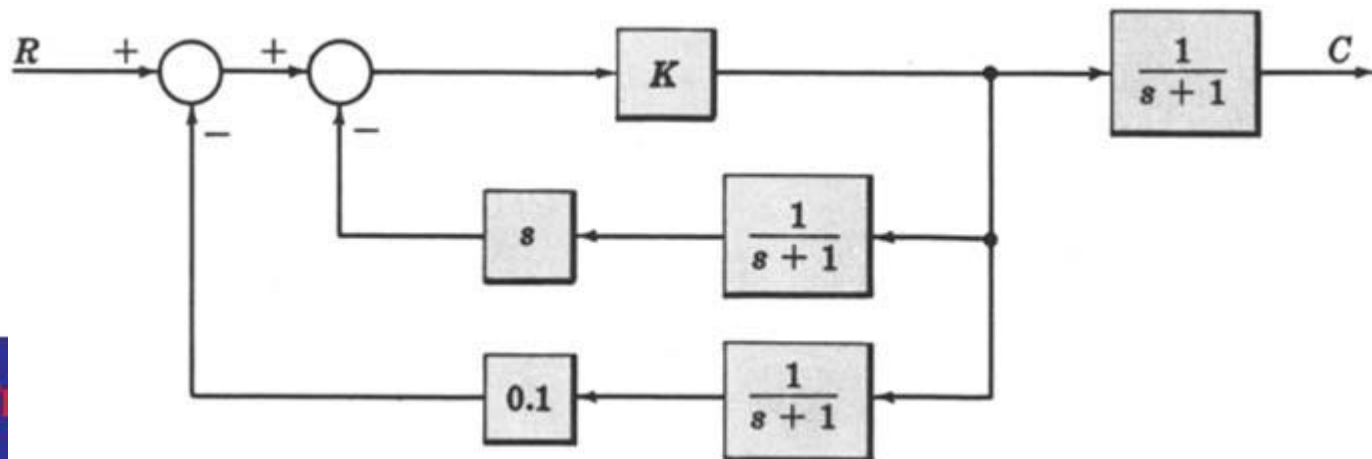
7. characteristic equation $1 + G(s)H(s) = 0$



8. closed loop poles and zeros if $K=10$.

Example-6

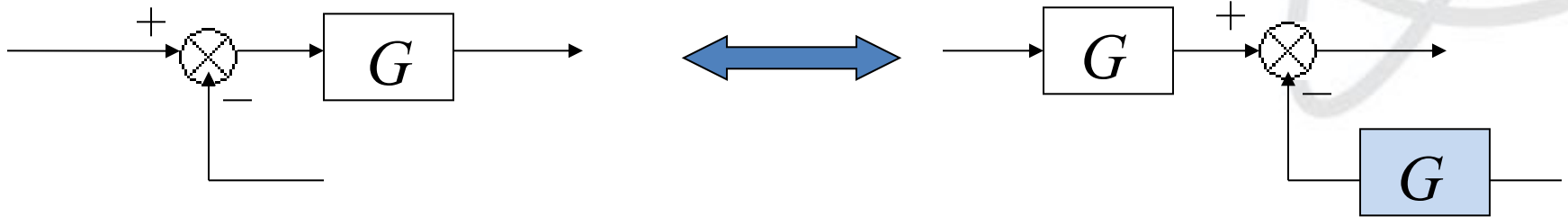
- For the system represented by the following block diagram determine:
 - Open loop transfer function
 - Feed Forward Transfer function
 - control ratio
 - feedback ratio
 - error ratio
 - closed loop transfer function
 - characteristic equation
 - closed loop poles and zeros if $K=100$.



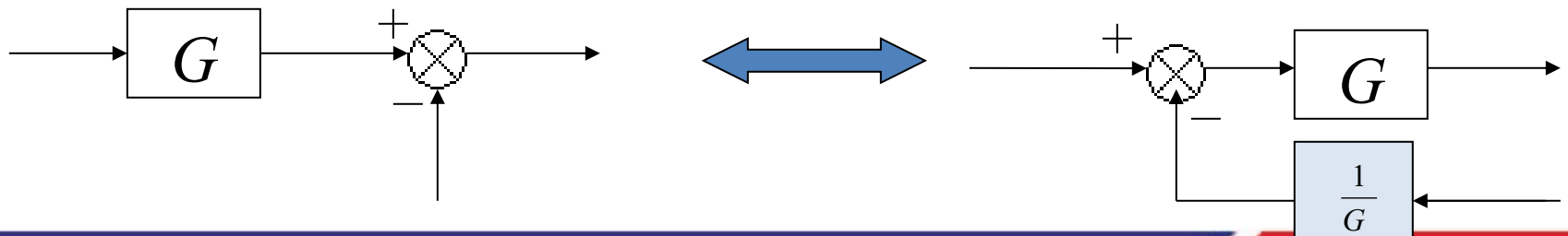


Reduction techniques

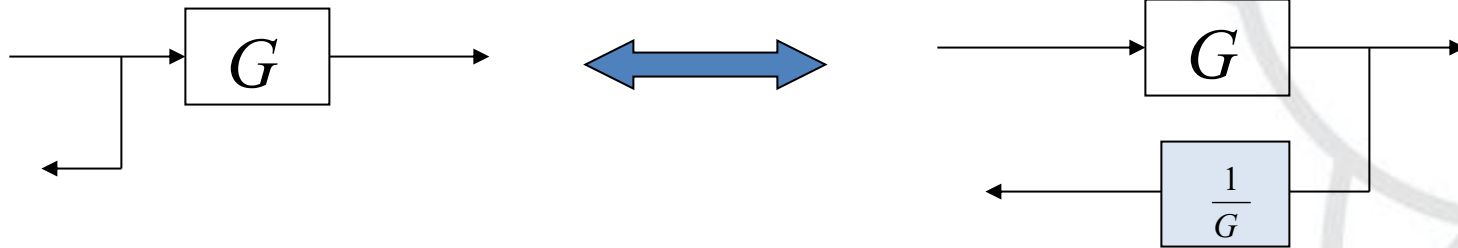
4. Moving a summing point behind a block



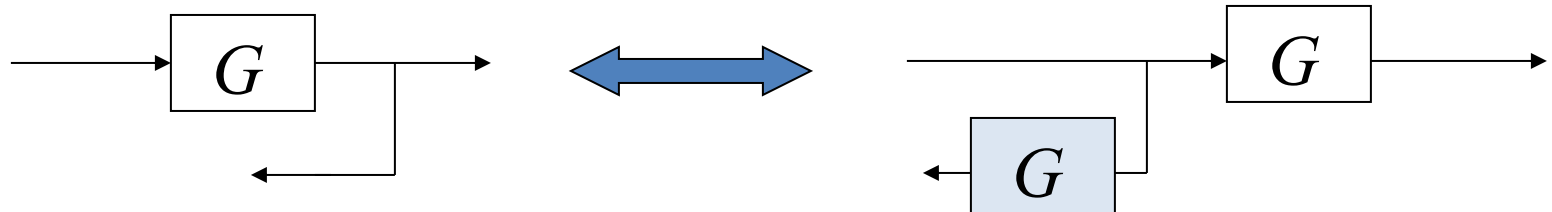
5. Moving a summing point ahead a block



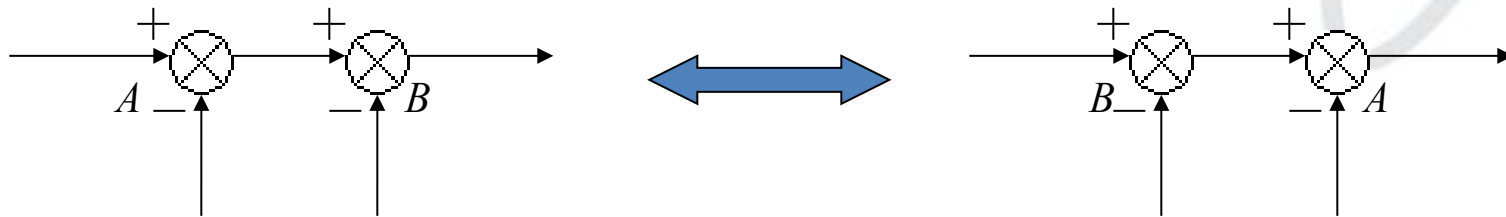
6. Moving a pickoff point behind a block



7. Moving a pickoff point ahead of a block

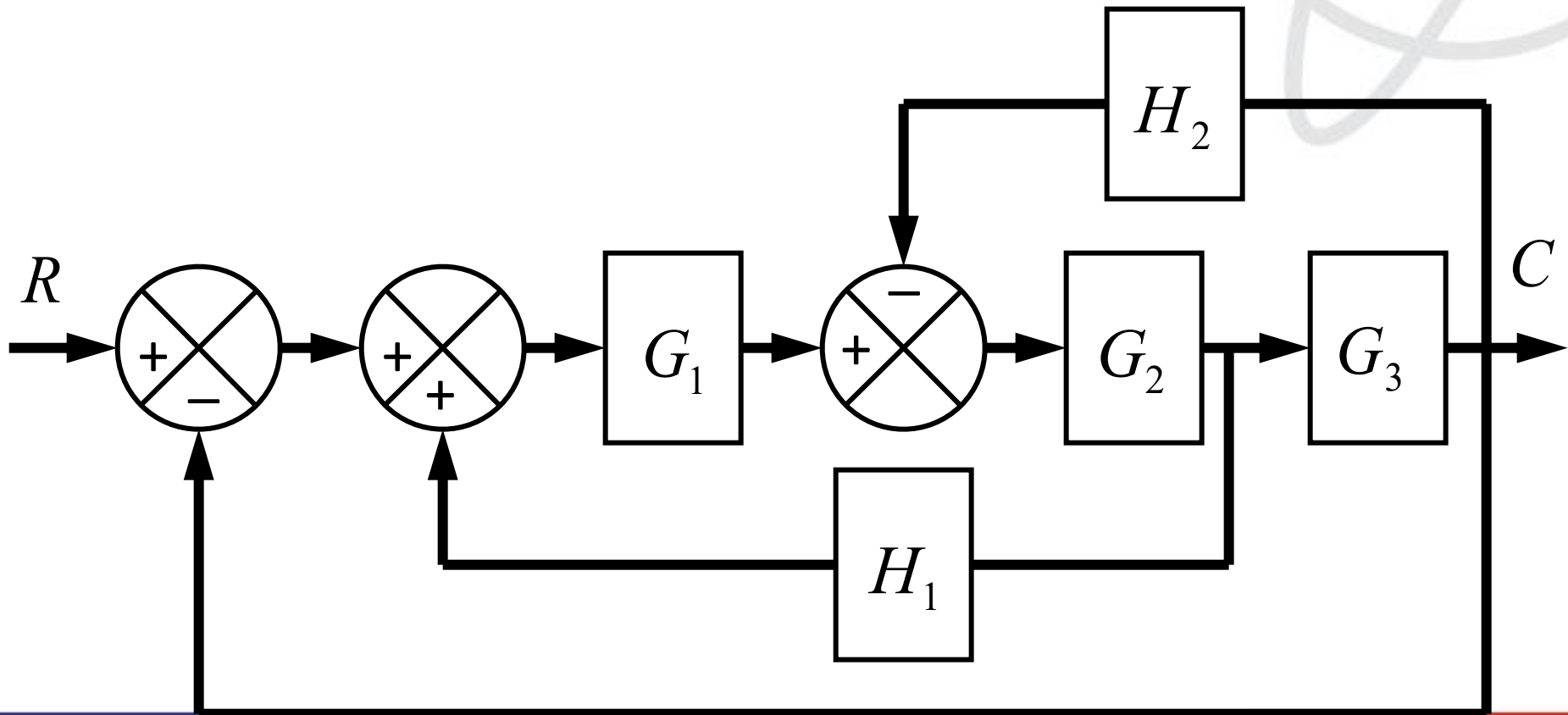


8. Swap with two neighboring summing points

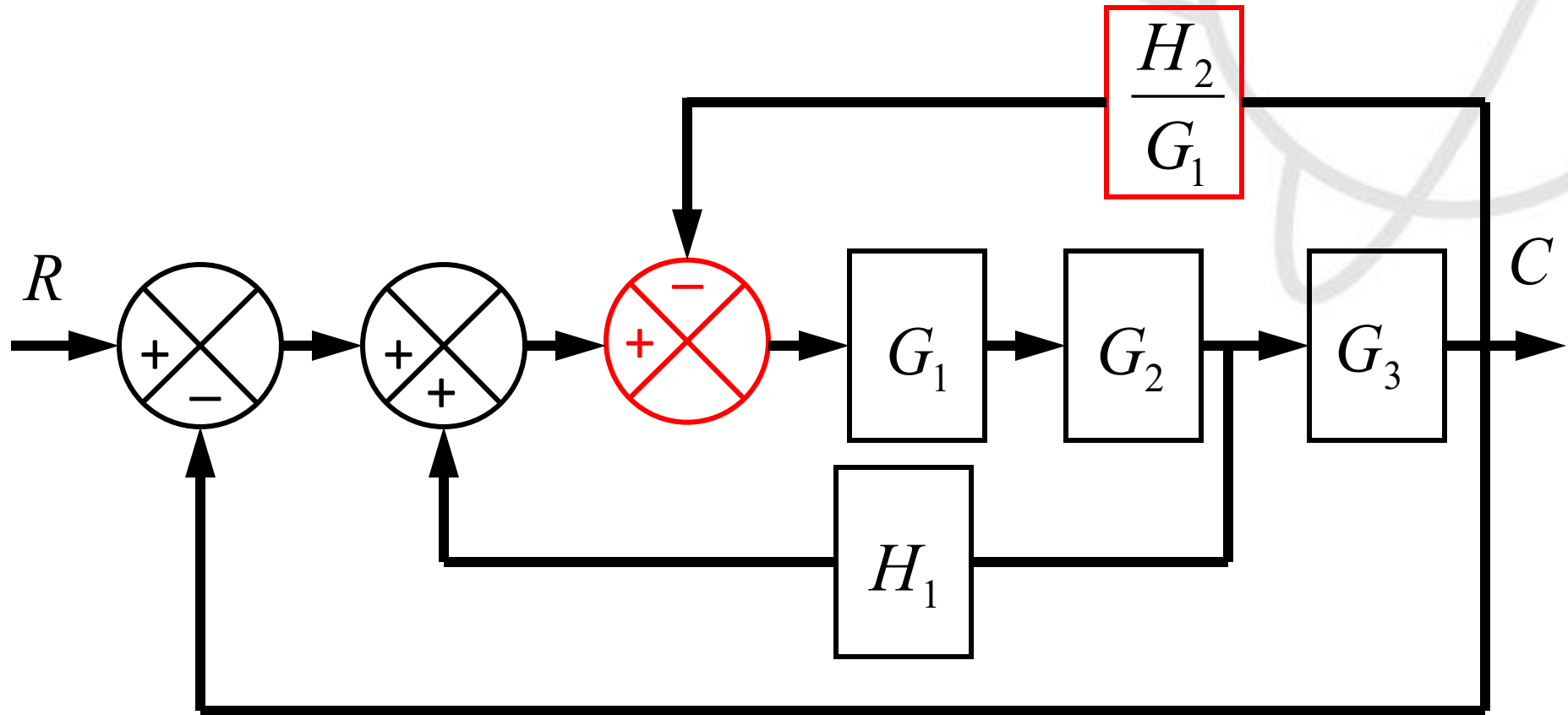


Example-7

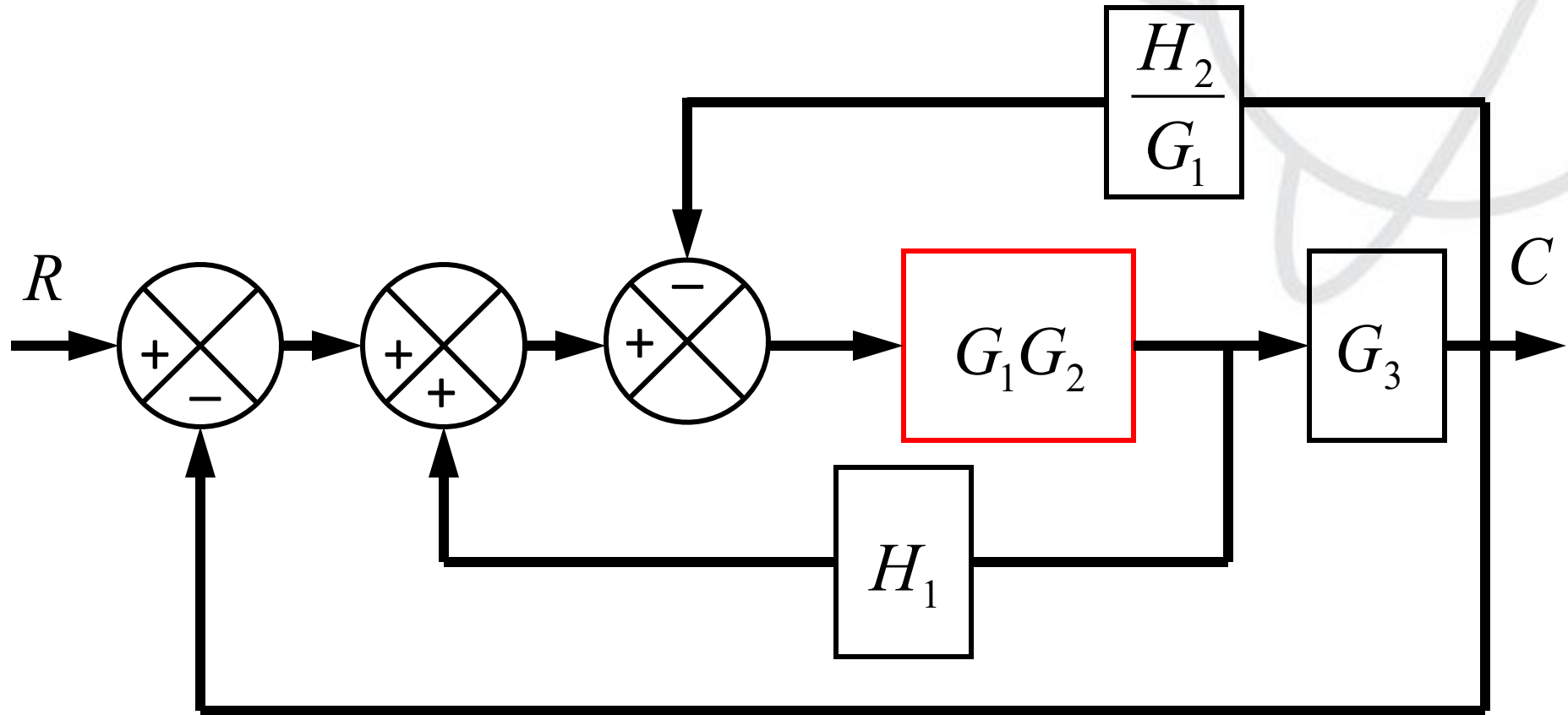
- Reduce the following block diagram to canonical form.



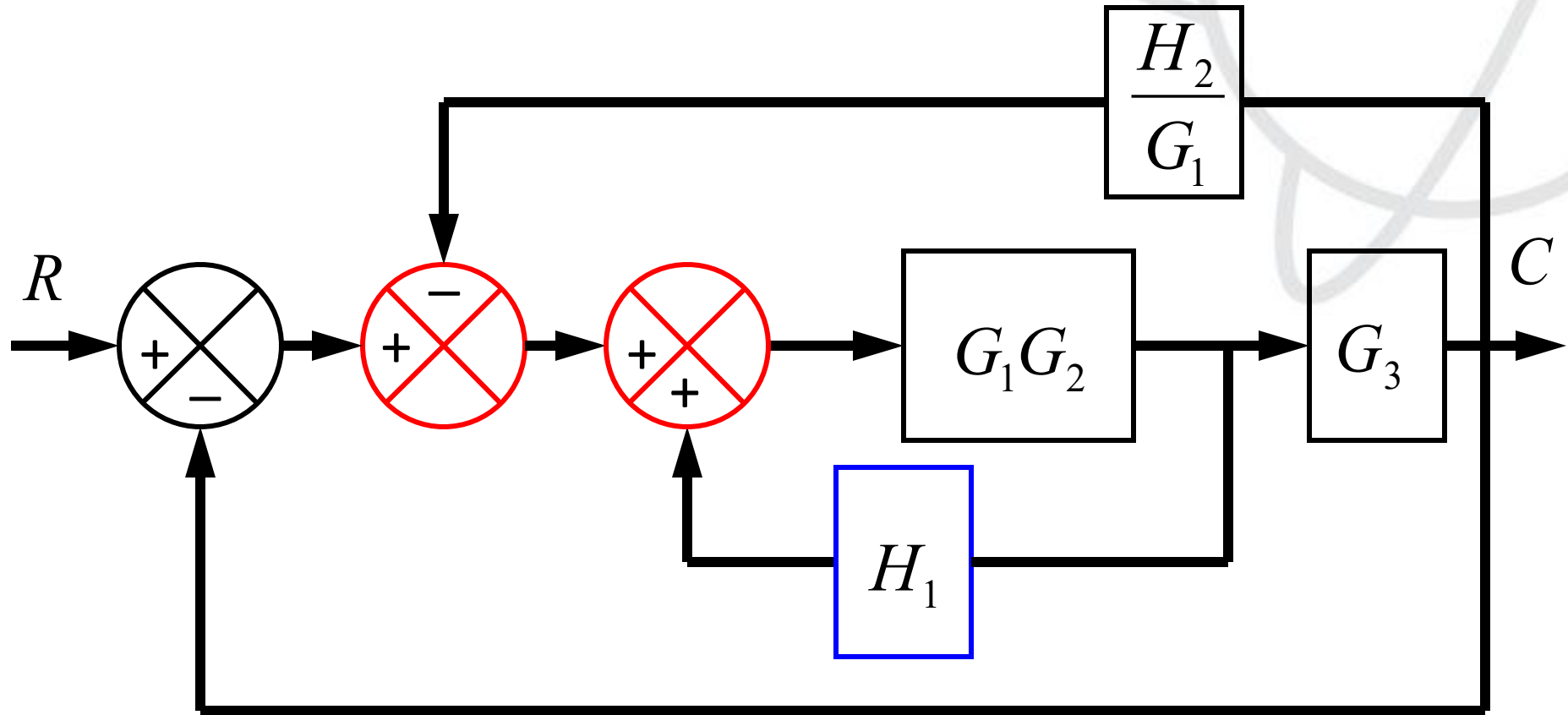
Example-7



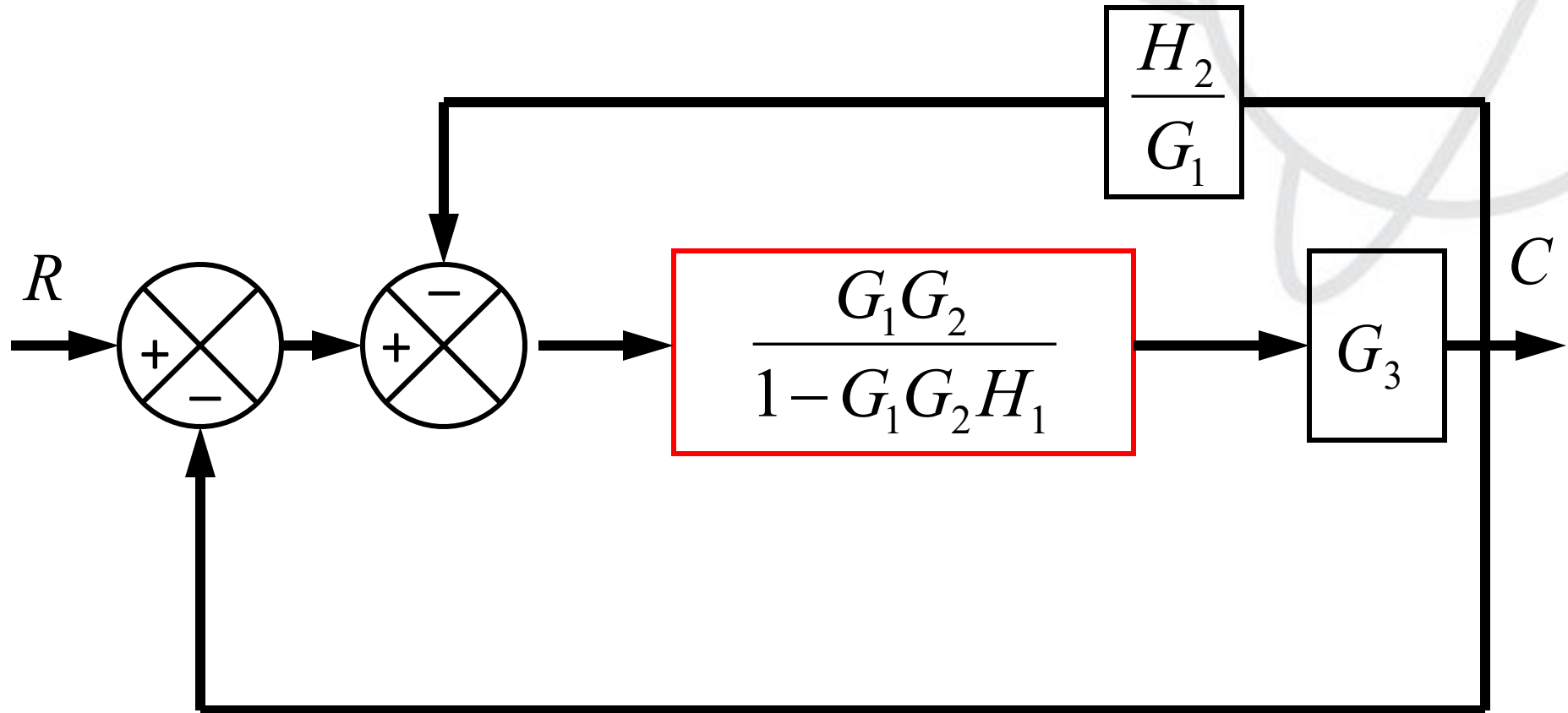
Example-7



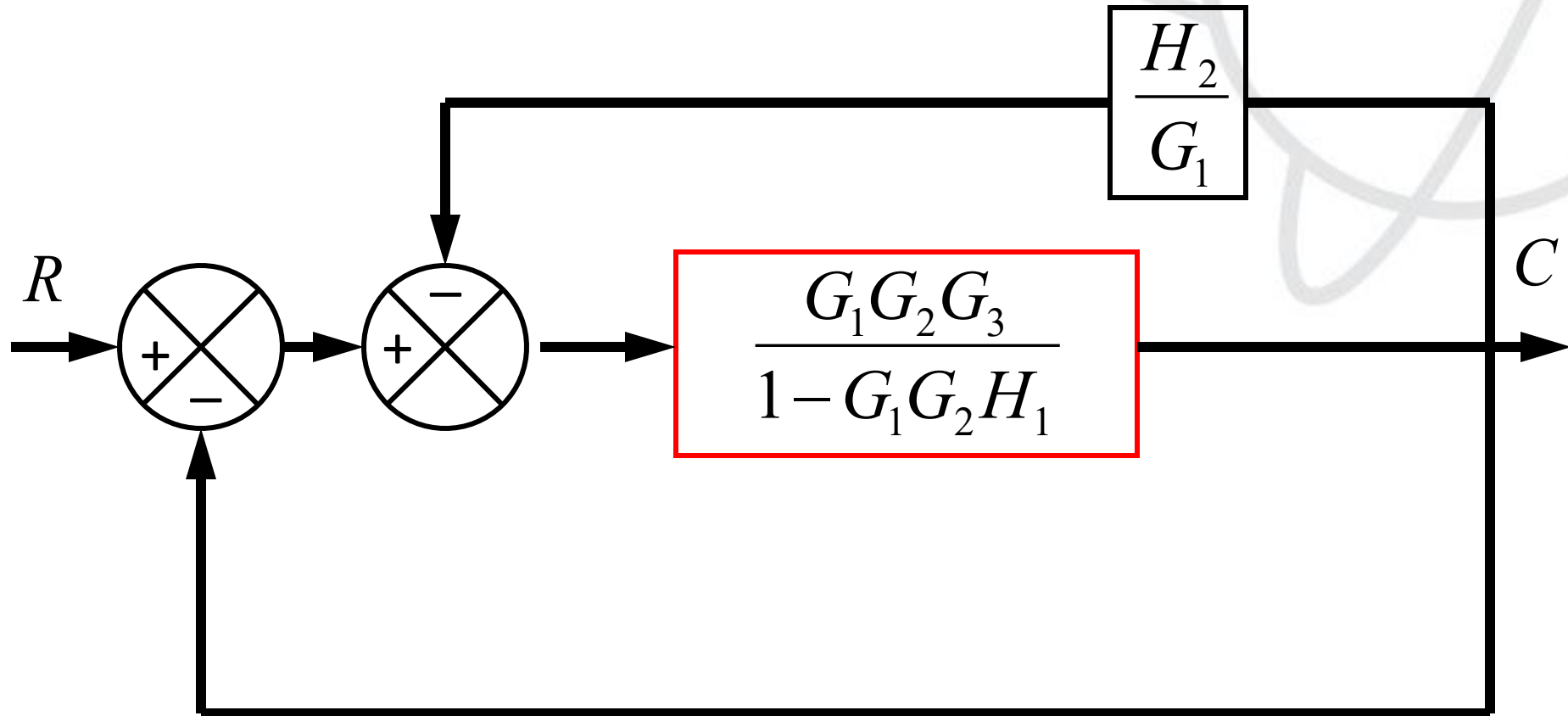
Example-7



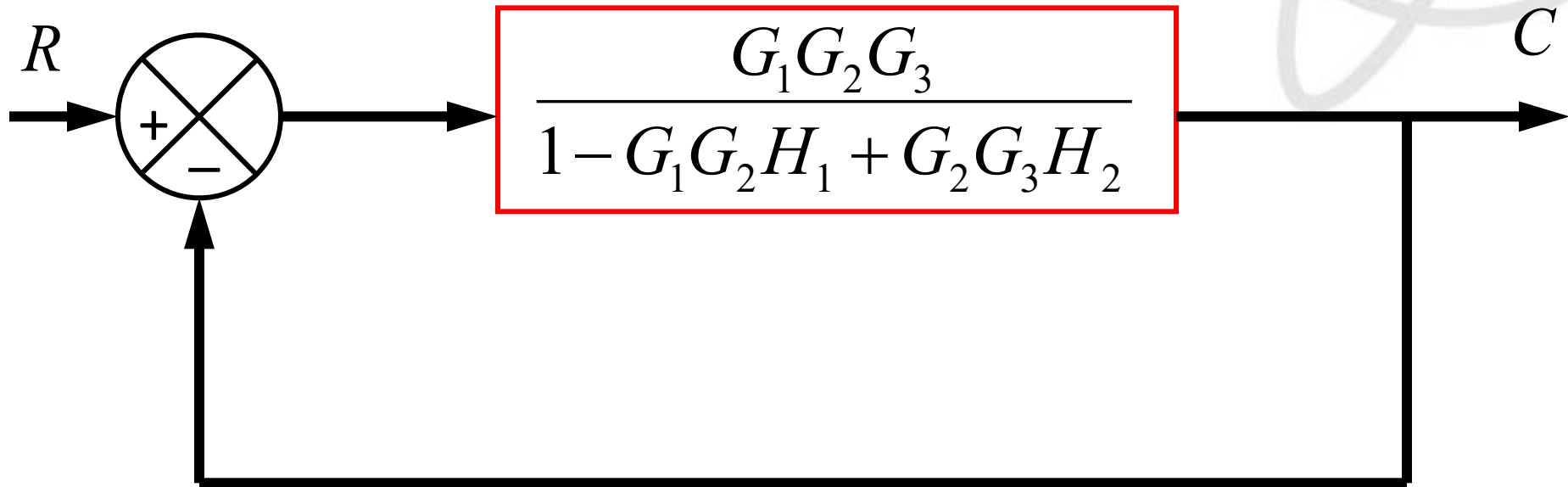
Example-7



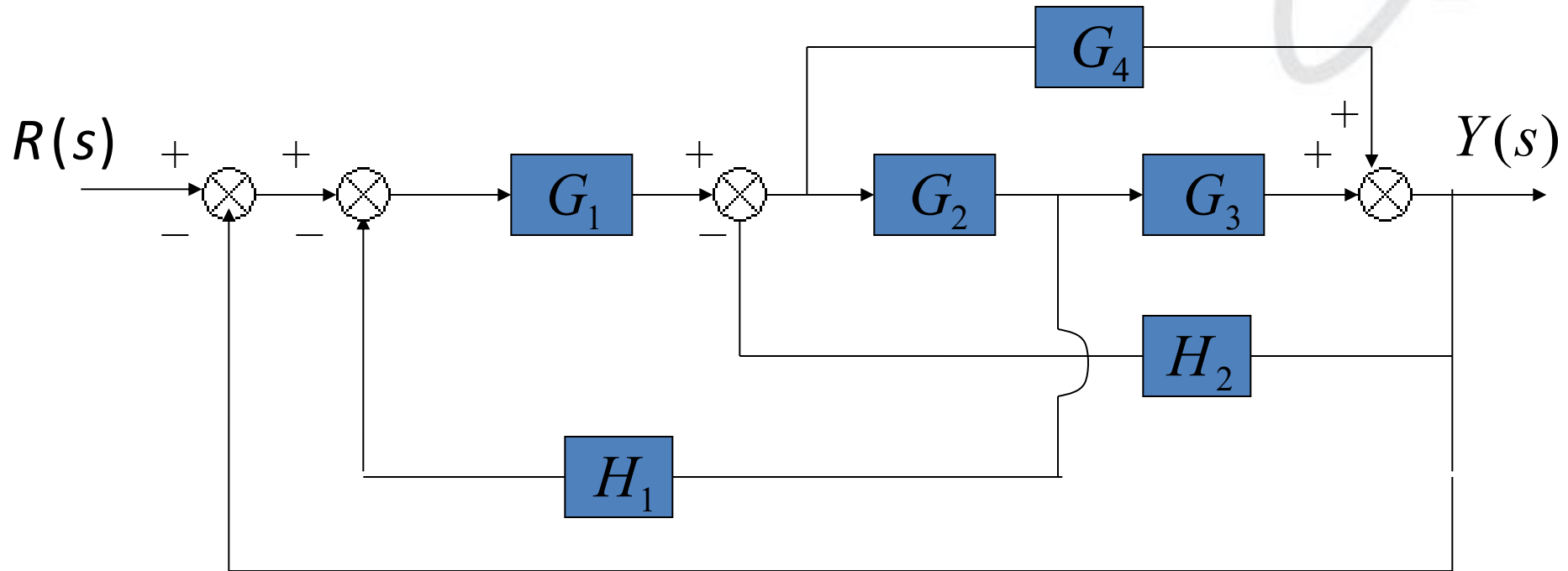
Example-7



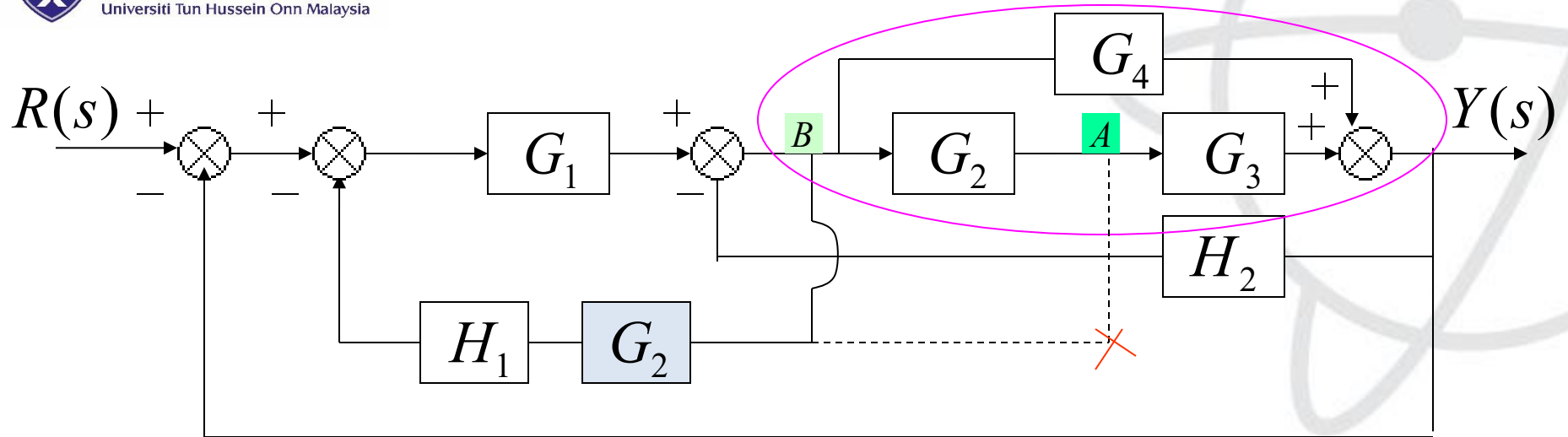
Example-7



Find the transfer function of the following block diagram

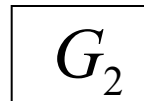


I

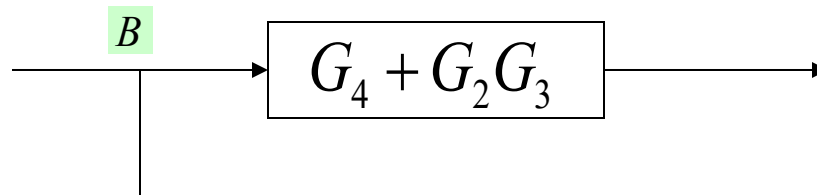


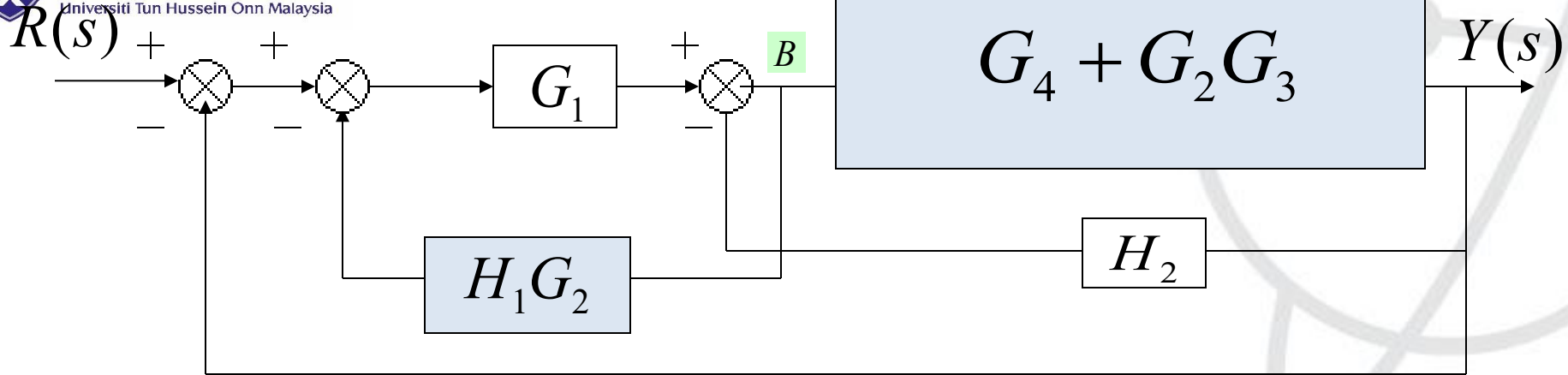
Solution:

1. Moving pickoff point A ahead of block G_2

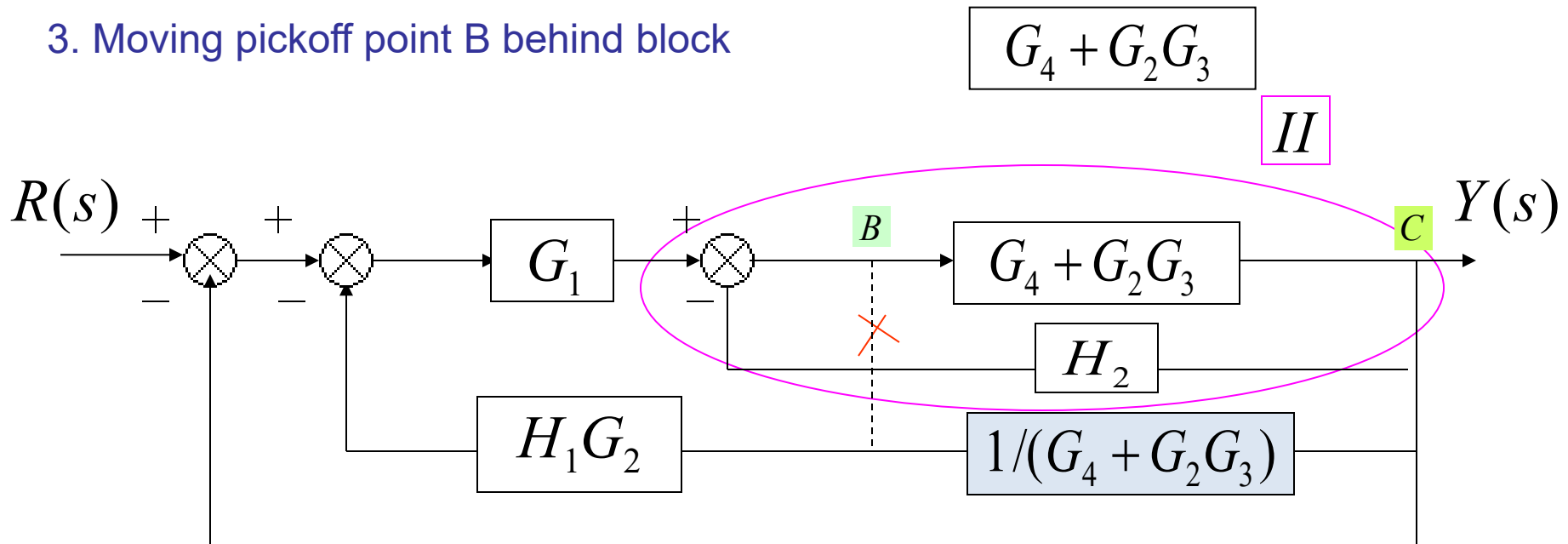


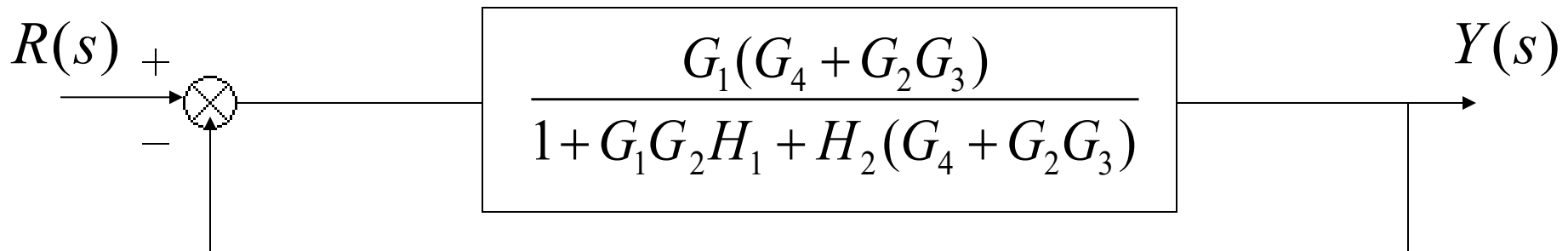
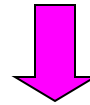
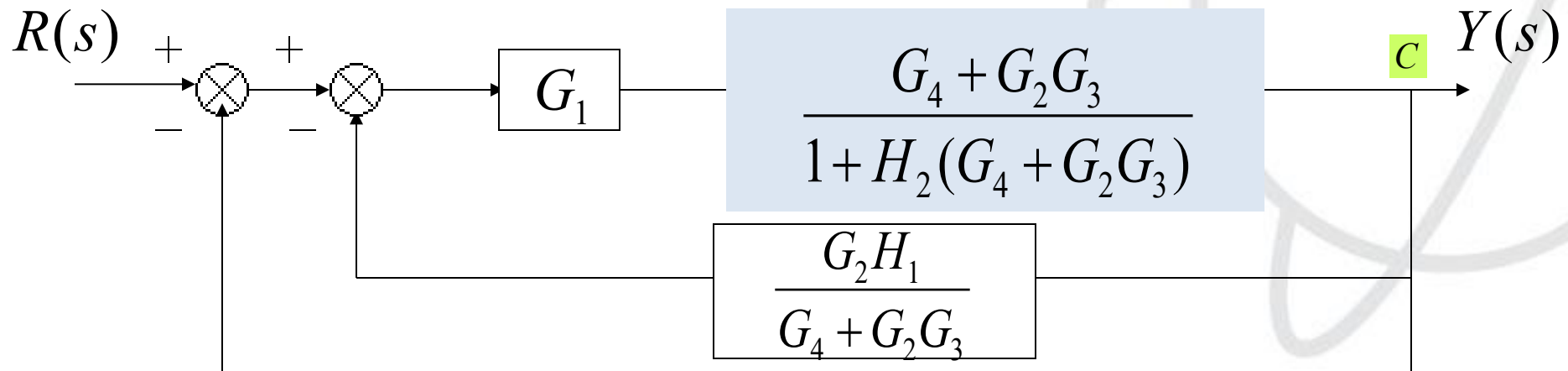
2. Eliminate loop I & simplify





3. Moving pickoff point B behind block

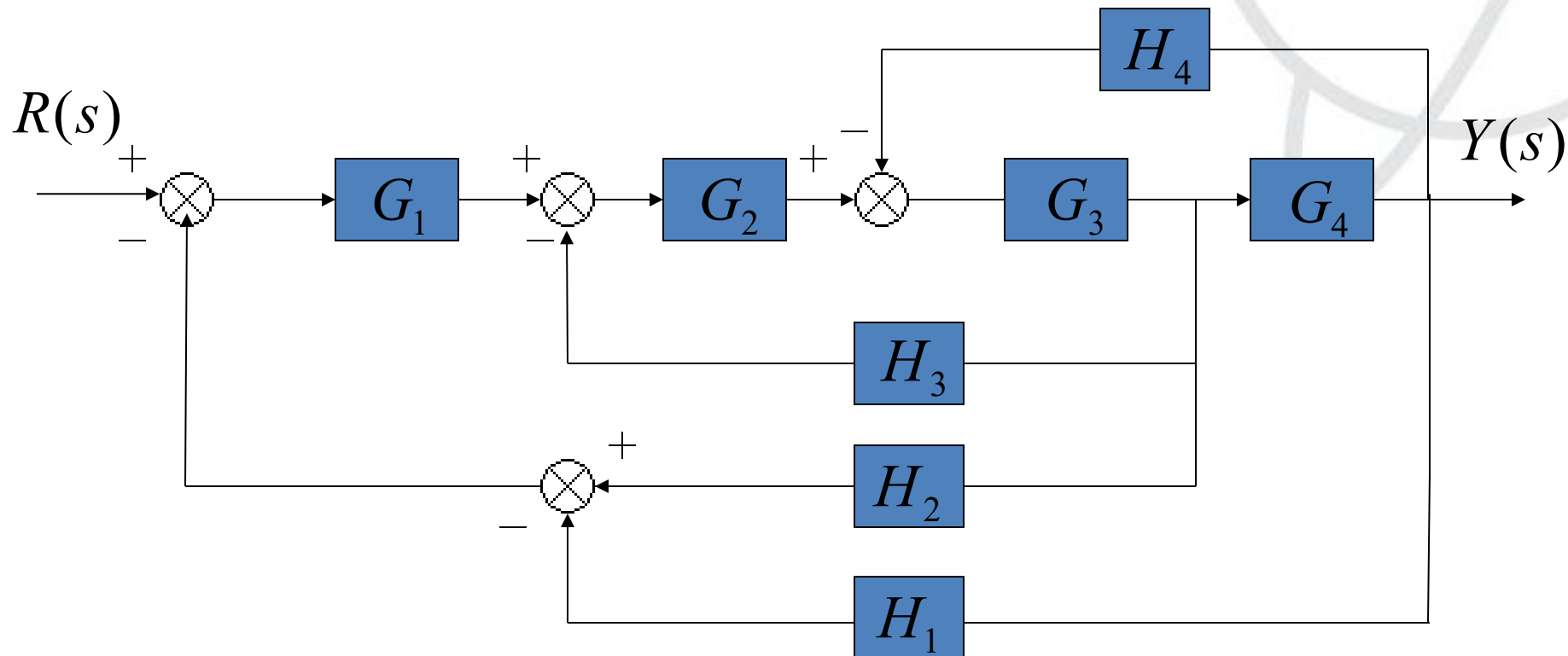




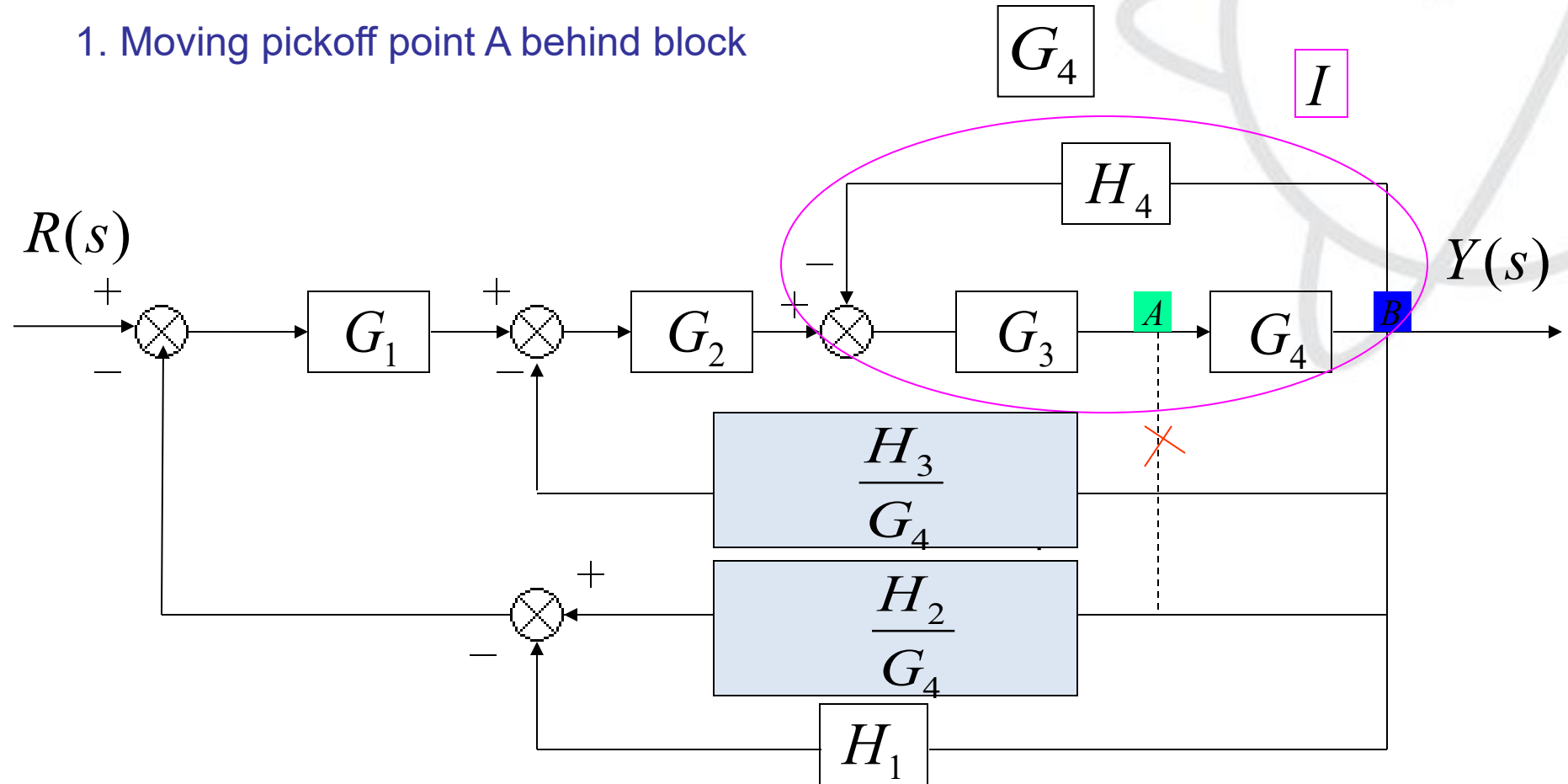
$$\frac{Y(s)}{R(s)} = \frac{G_1(G_4 + G_2 G_3)}{1 + G_1 G_2 H_1 + H_2(G_4 + G_2 G_3) + G_1(G_4 + G_2 G_3)}$$

Example 9

Find the transfer function of the following block diagrams

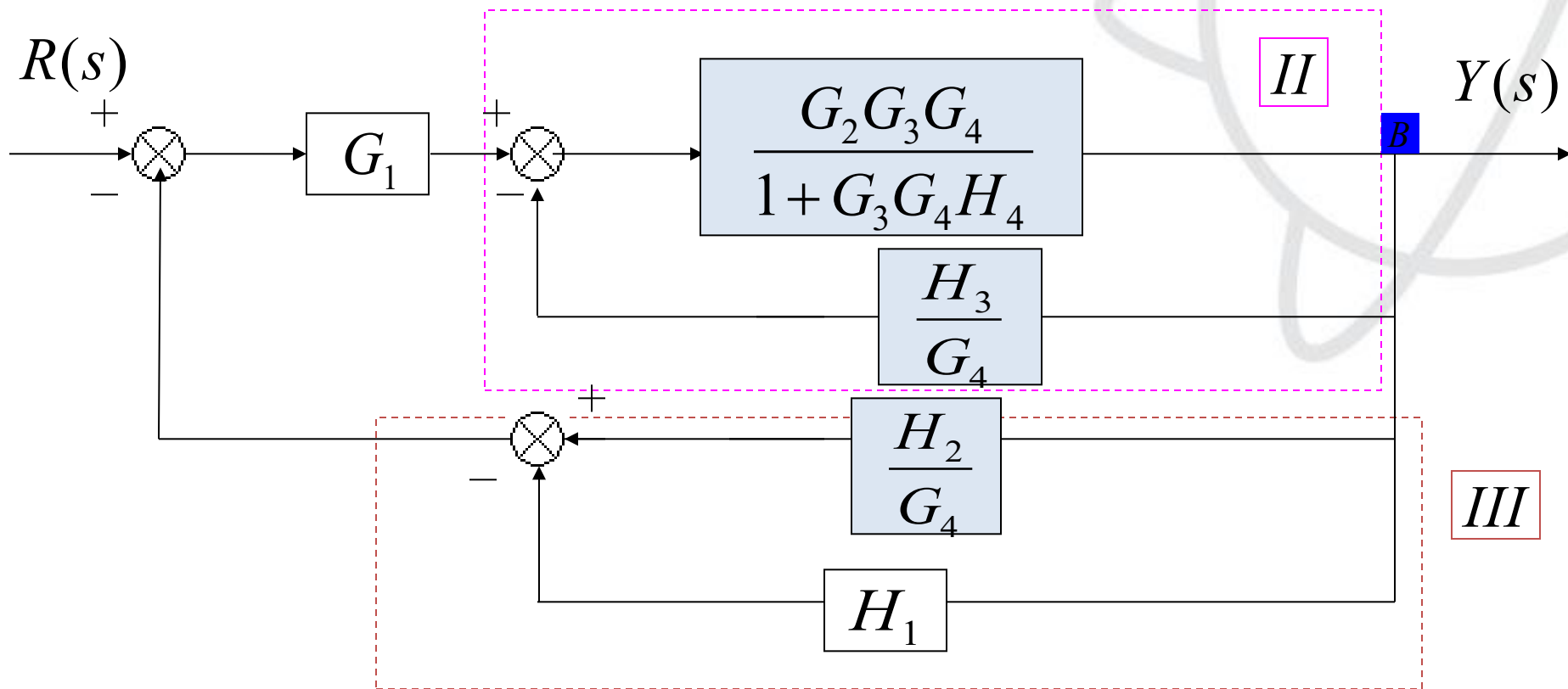


1. Moving pickoff point A behind block

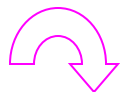




2. Eliminate loop I and Simplify



II



feedback

$$\frac{G_2 G_3 G_4}{1 + G_3 G_4 H_4 + G_2 G_3 H_3}$$

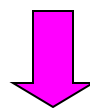
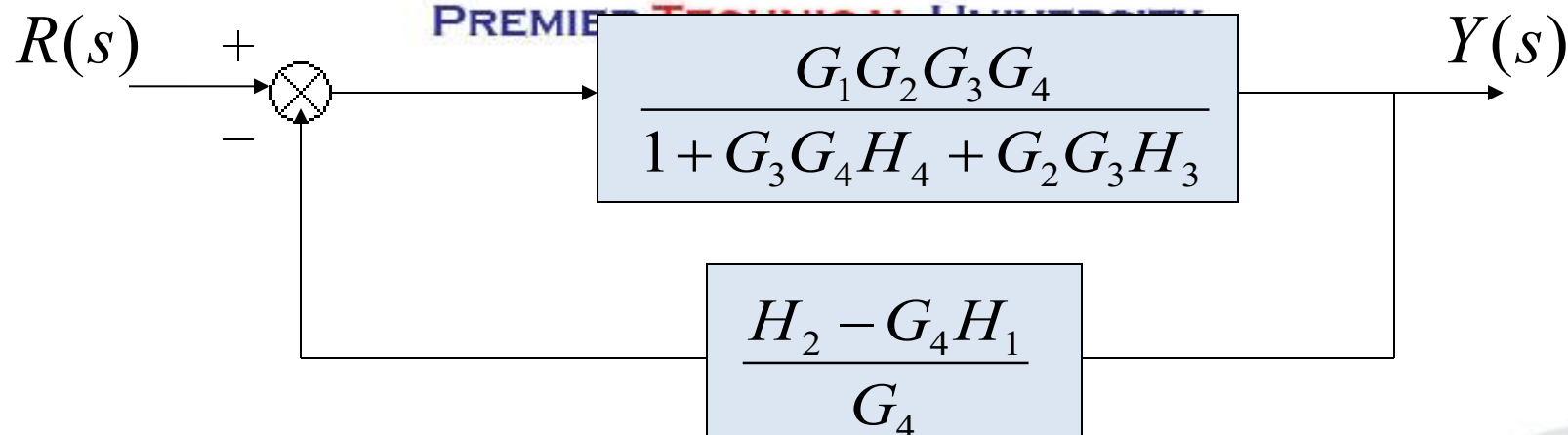
III



Not feedback

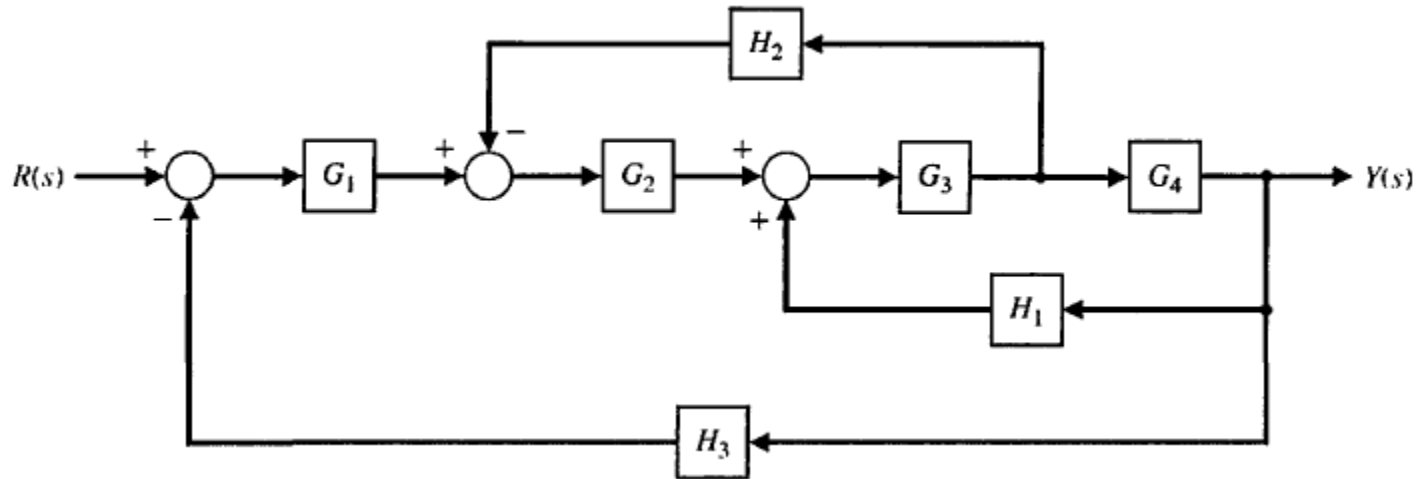
$$\frac{H_2 - G_4 H_1}{G_4}$$

3. Eliminate loop II & III

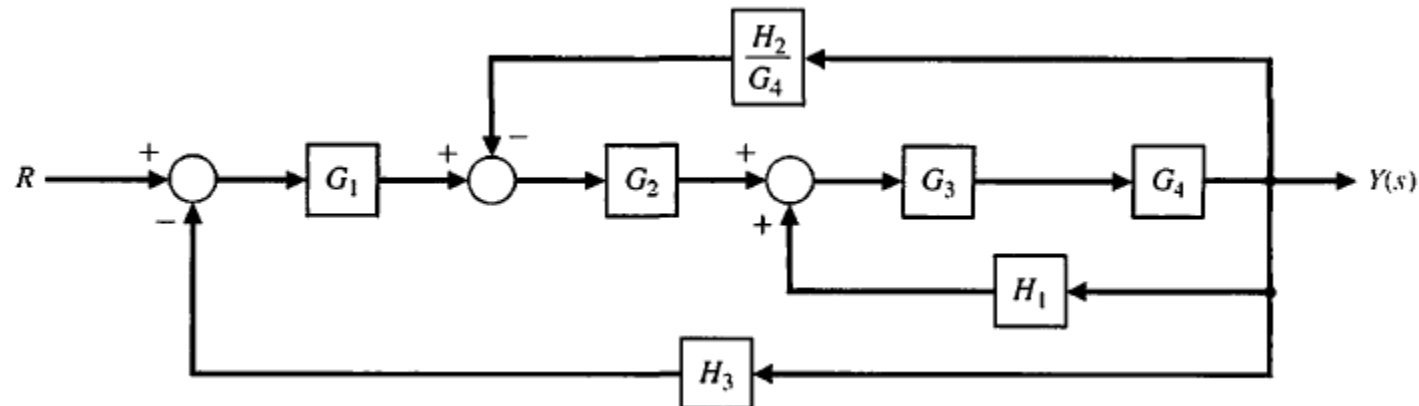


$$\frac{Y(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4}{1 + G_2 G_3 H_3 + G_3 G_4 H_4 + G_1 G_2 G_3 H_2 - G_1 G_2 G_3 G_4 H_1}$$

Example-10: Reduce the Block Diagram.

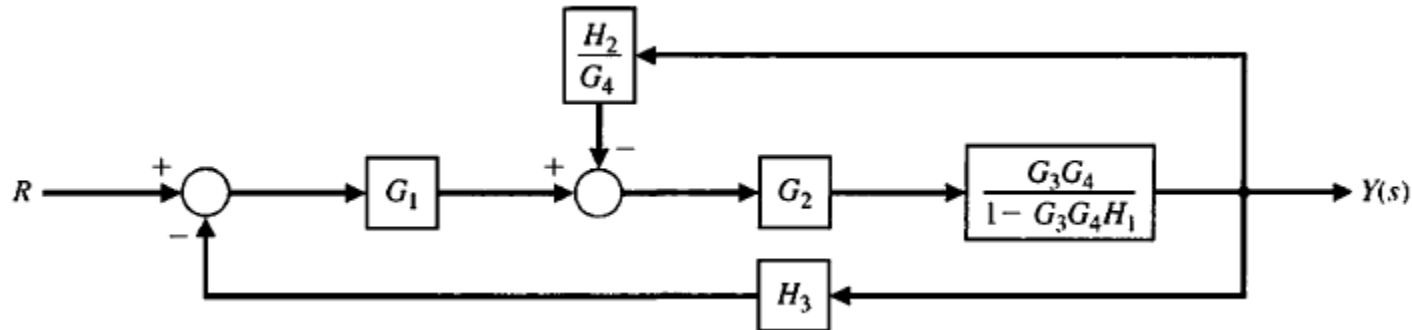


First, to eliminate the loop $G_3G_4H_1$, we move H_2 behind block G_4

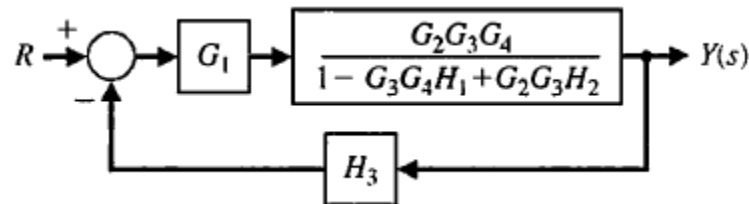


Example-10: Continue.

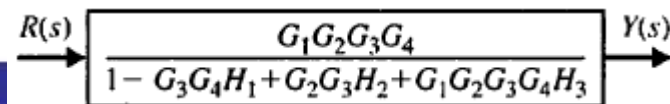
Eliminating the loop $G_3G_4H_1$ we obtain



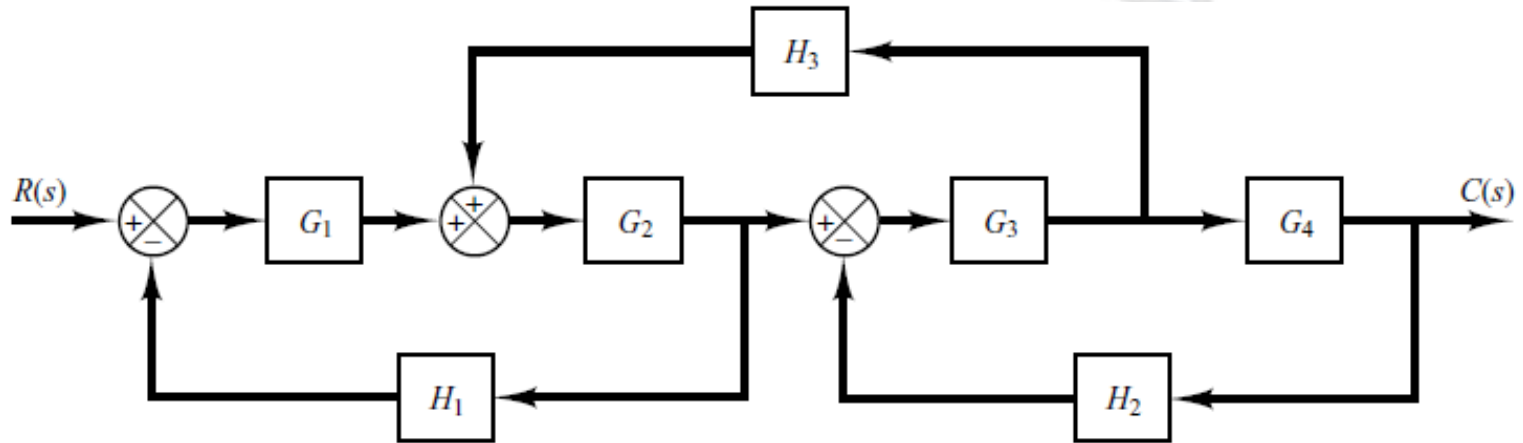
Then, eliminating the inner loop containing H_2/G_4 , we obtain



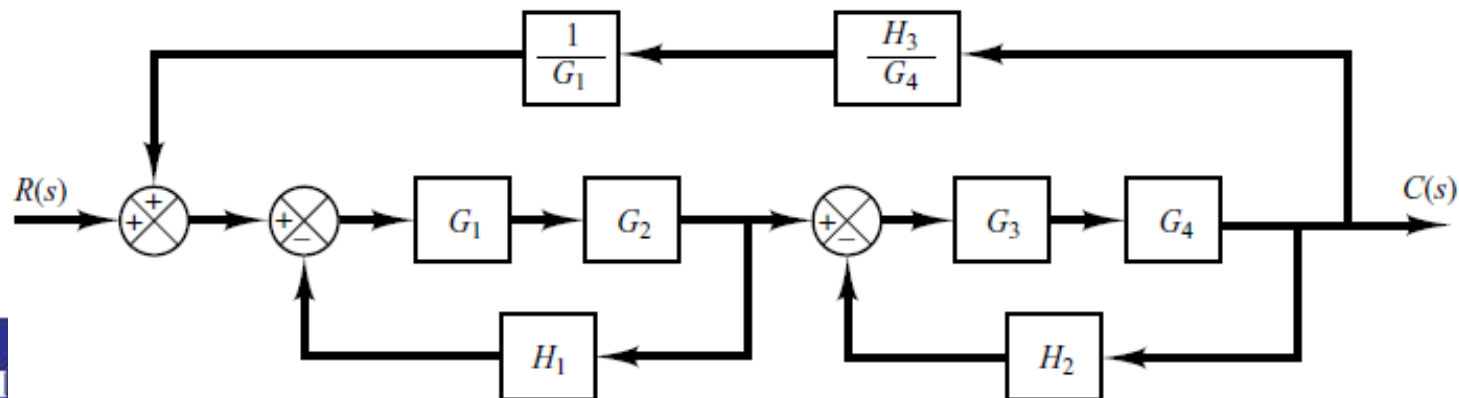
Finally, by reducing the loop containing H_3 , we obtain



Example-11: Simplify the block diagram then obtain the close-loop transfer function $C(S)/R(S)$. (from Ogata: Page-47)

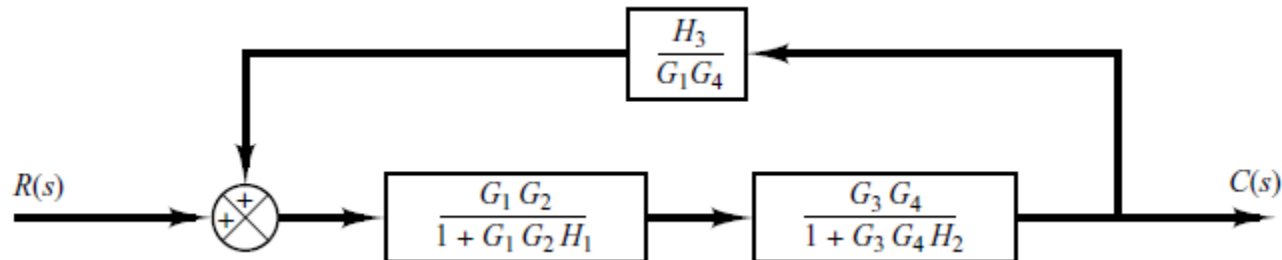


First move the branch point between G_3 and G_4 to the right-hand side of the loop containing G_3 , G_4 , and H_2 . Then move the summing point between G_1 and G_2 to the left-hand side of the first summing point.

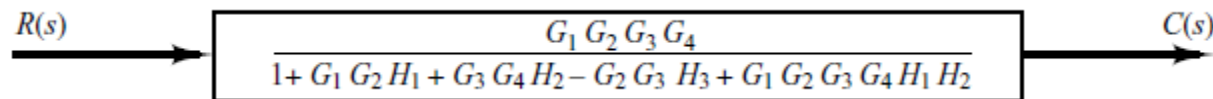


Example-11: Continue.

By simplifying each loop, the block diagram can be modified as



Further simplification results in



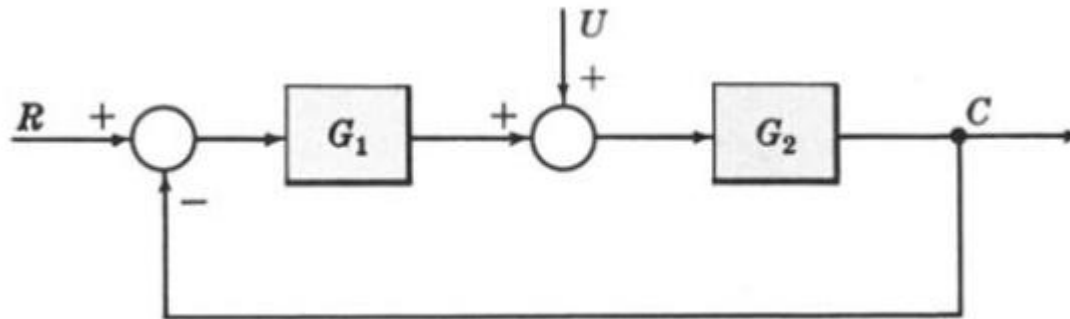
the closed-loop transfer function $C(s)/R(s)$ is obtained as

$$\frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4}{1 + G_1 G_2 H_1 + G_3 G_4 H_2 - G_2 G_3 H_3 + G_1 G_2 G_3 G_4 H_1 H_2}$$

Superposition of Multiple Inputs

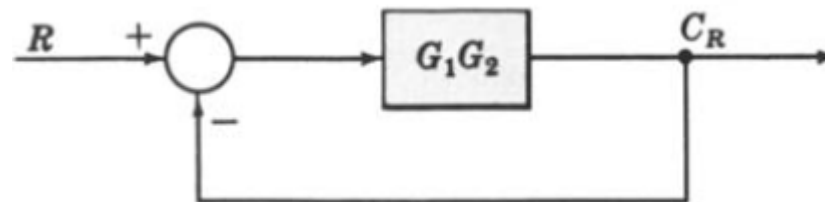
- Step 1:** Set all inputs except one equal to zero.
- Step 2:** Transform the block diagram to canonical form, using the transformations of Section 7.5.
- Step 3:** Calculate the response due to the chosen input acting alone.
- Step 4:** Repeat Steps 1 to 3 for each of the remaining inputs.
- Step 5:** Algebraically add all of the responses (outputs) determined in Steps 1 to 4. This sum is the total output of the system with all inputs acting simultaneously.

Example-12: Multiple Input System. Determine the output C due to inputs R and U using the Superposition Method.



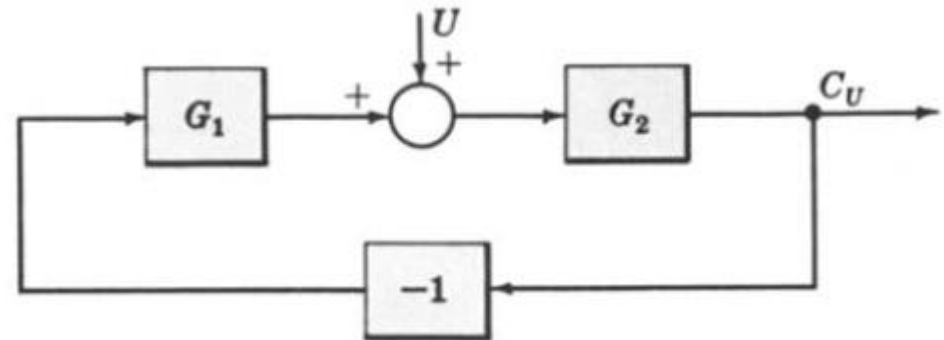
Step 1: Put $U \equiv 0$.

Step 2: The system reduces to



Step 3: the output C_R due to input R is $C_R = [G_1G_2/(1 + G_1G_2)]R$.

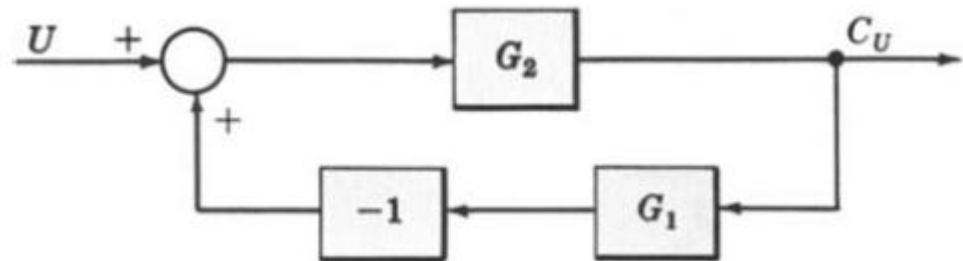
Example-12: Continue.



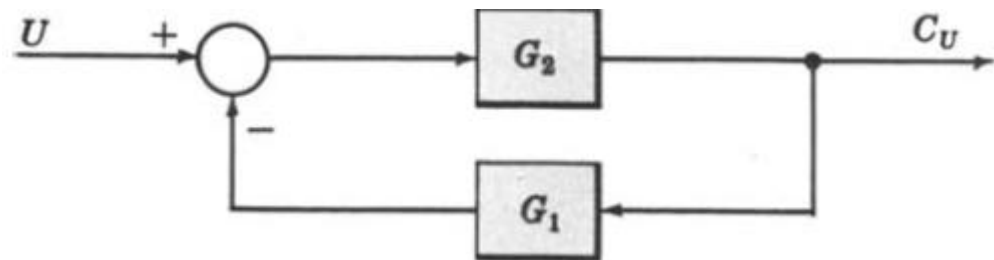
Step 4a: Put $R = 0$.

Step 4b: Put -1 into a block, representing the negative feedback effect:

Rearrange the block diagram:



Let the -1 block be absorbed into the summing point:



Step 4c: the output C_U due to input U is $C_U = [G_2 / (1 + G_1 G_2)] U$.

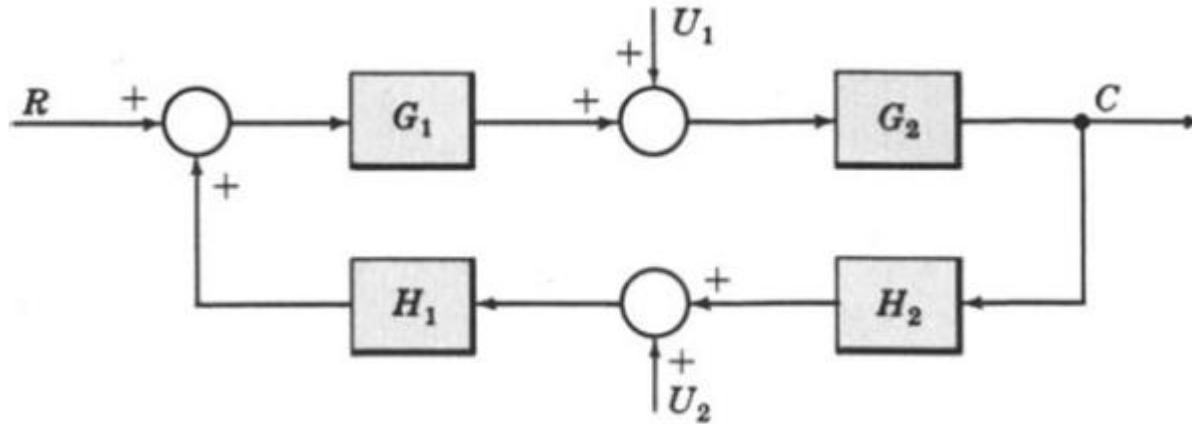
Example-12: Continue.

Step 5: The total output is $C = C_R + C_U$

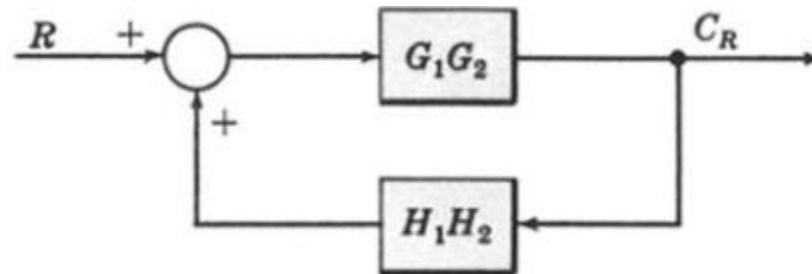
$$= \left[\frac{G_1 G_2}{1 + G_1 G_2} \right] R + \left[\frac{G_2}{1 + G_1 G_2} \right] U$$

$$= \left[\frac{G_2}{1 + G_1 G_2} \right] [G_1 R + U]$$

Example-13: **Multiple-Input System**. Determine the output C due to inputs R , U_1 and U_2 using the Superposition Method.



Let $U_1 = U_2 = 0$.

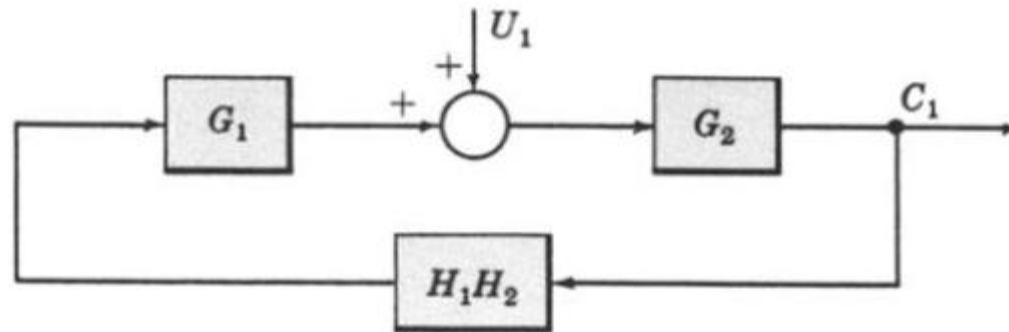


$$C_R = [G_1 G_2 / (1 - G_1 G_2 H_1 H_2)] R$$

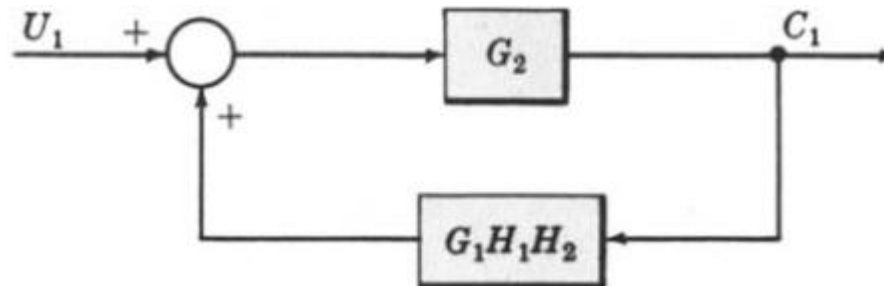
where C_R is the output due to R acting alone.

Example-13: Continue.

Now let $R = U_2 = 0$.



Rearranging the blocks, we get

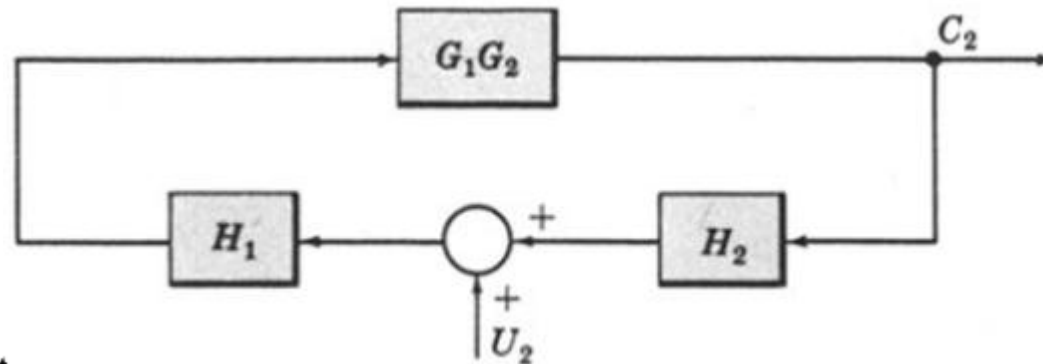


$$C_1 = [G_2 / (1 - G_1 G_2 H_1 H_2)] U_1$$

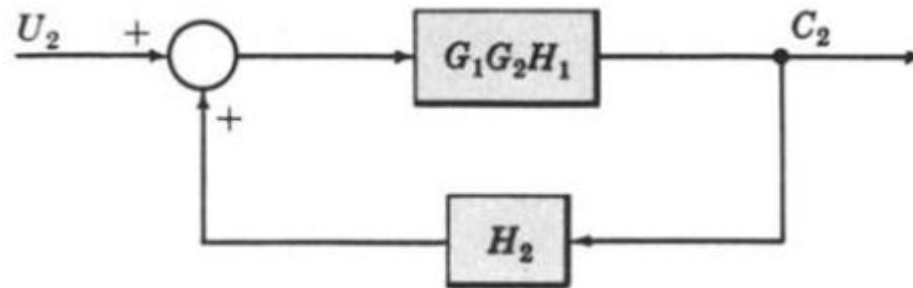
where C_1 is the response due to U_1 acting alone.

Example-13: Continue.

Finally, let $R = U_1 = 0$.



Rearranging the blocks, we get



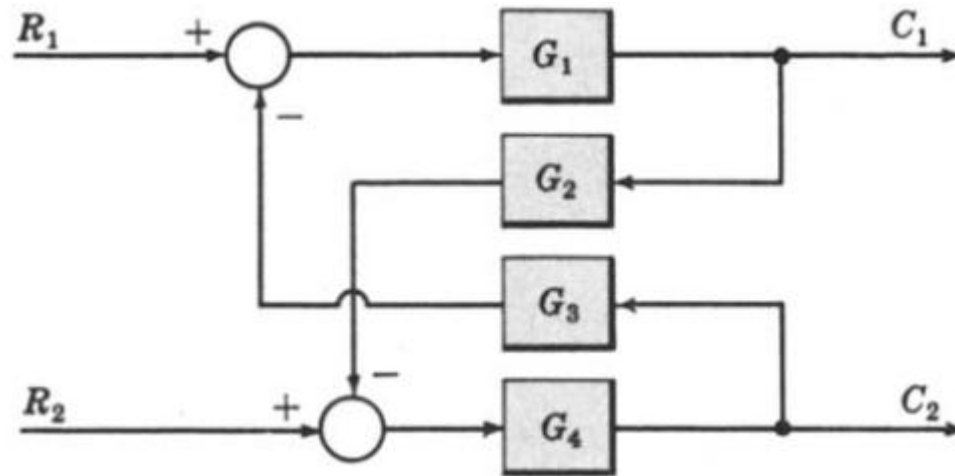
$$C_2 = [G_1G_2H_1 / (1 - G_1G_2H_1H_2)]U_2$$

where C_2 is the response due to U_2 acting alone.

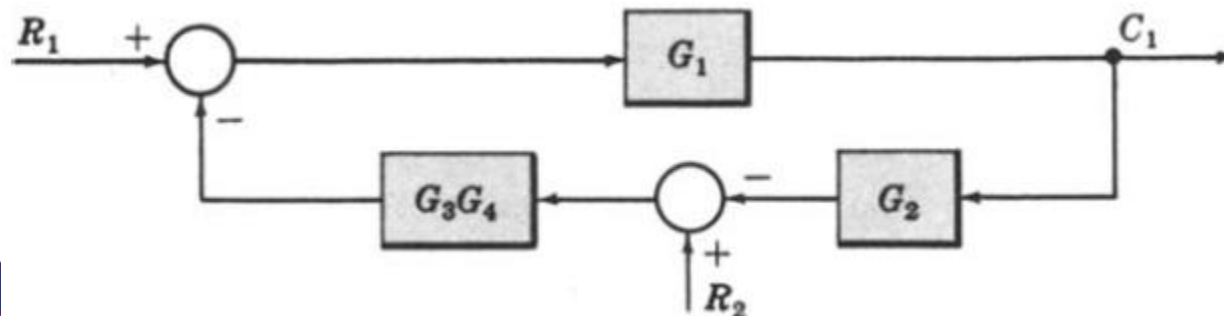
By superposition, the total output is

$$C = C_R + C_1 + C_2 = \frac{G_1G_2R + G_2U_1 + G_1G_2H_1U_2}{1 - G_1G_2H_1H_2}$$

Example-14: Multi-Input Multi-Output System. Determine C_1 and C_2 due to R_1 and R_2 .



First ignoring the output C_2 .



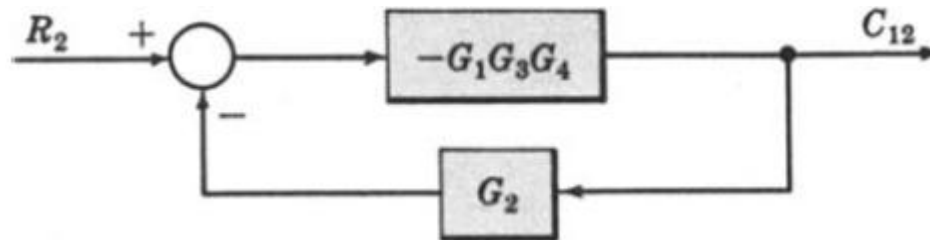
Example-14: Continue.

Letting $R_2 = 0$ and combining the summing points,



Hence C_{11} , the output at C_1 due to R_1 alone, is $C_{11} = G_1 R_1 / (1 - G_1 G_2 G_3 G_4)$.

For $R_1 = 0$,

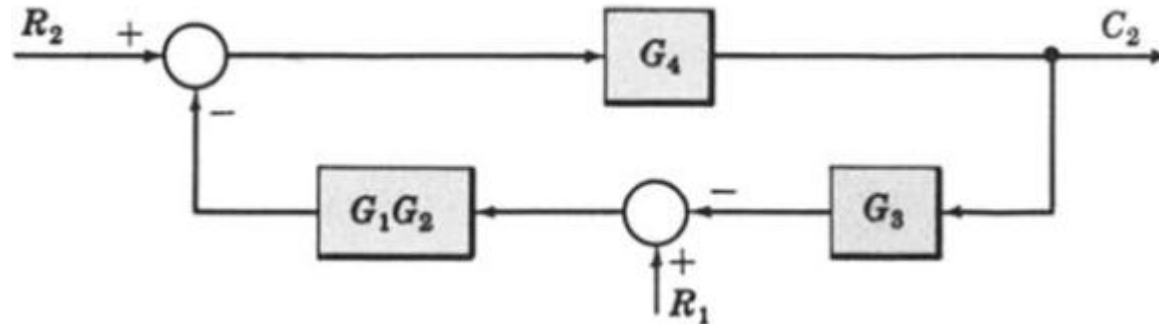


Hence $C_{12} = -G_1 G_3 G_4 R_2 / (1 - G_1 G_2 G_3 G_4)$ is the output at C_1 due to R_2 alone.

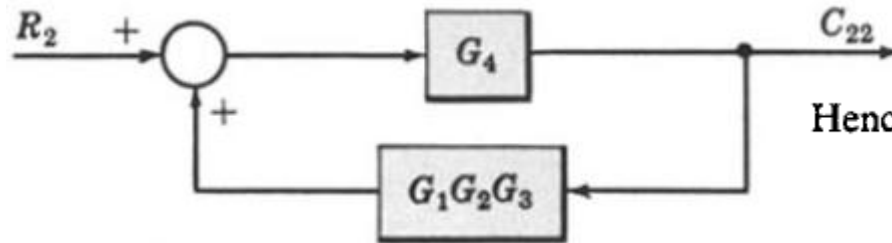
Thus $C_1 = C_{11} + C_{12} = (G_1 R_1 - G_1 G_3 G_4 R_2) / (1 - G_1 G_2 G_3 G_4)$

Example-14: Continue.

Now we reduce the original block diagram, ignoring output C_1 .

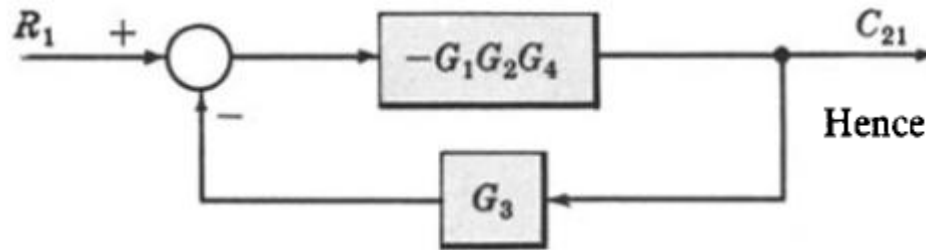


When $R_1 = 0$,



$$\text{Hence } C_{22} = G_4 R_2 / (1 - G_1 G_2 G_3 G_4)$$

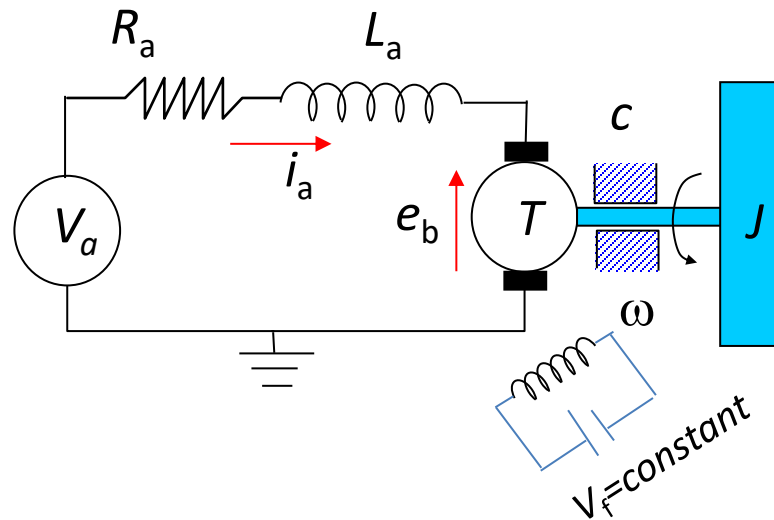
When $R_2 = 0$,



$$\text{Hence } C_{21} = -G_1 G_2 G_4 R_1 / (1 - G_1 G_2 G_3 G_4)$$

$$\text{Finally, } C_2 = C_{22} + C_{21} = (G_4 R_2 - G_1 G_2 G_4 R_1) / (1 - G_1 G_2 G_3 G_4)$$

Block Diagram of Armature Controlled D.C Motor

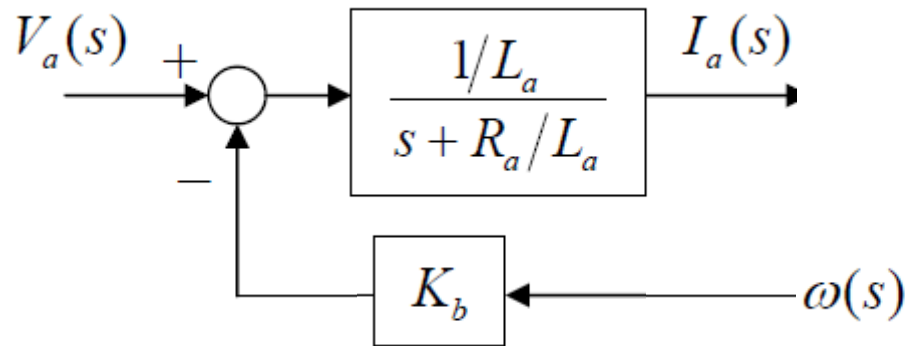


$$(L_a s + R_a) I_a(s) + K_b \omega(s) = V_a(s)$$

$$(J s + c) \omega(s) = K_m I_a(s)$$

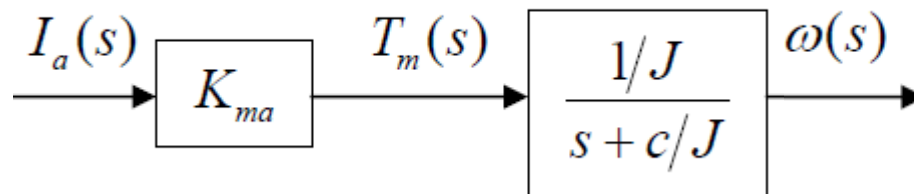
Block Diagram of Armature Controlled D.C Motor

$$(L_a s + R_a) I_a(s) + K_b \omega(s) = V_a(s)$$



Block Diagram of Armature Controlled D.C Motor

$$(Js + c)\omega(s) = K_{ma}I_a(s)$$



Block Diagram of Armature Controlled D.C Motor

